Assessment of the Walleye pollock Stock in the Eastern Bering Sea

Jim Ianelli Carey McGilliard Sophia Wassermann

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Executive Summary

This document presents the 2025 Eastern Bering Sea (EBS) walleye pollock assessment in the ASAR framework, using the same underlying assessment outputs developed in the source EBS pollock project.

Assessment Model

The base model remains the established ADMB implementation used in recent AFSC cycles. For this ASAR report, base-model outputs were standardized into the `out_new` structure from ADMB report products, and complementary cross-model diagnostics were retained from the 2025 comparison workflow.

Stock Status and Benchmarks

The terminal assessment year in this report is 2024. Key terminal quantities are:

- Spawning biomass (`SBend`): 3,389.48
- Fishing mortality (`Fend`): 0.1449
- Reference fishing mortality (`Ftarg`): 0.499
- Relative fishing pressure (`F/Ftarg`): 0.29

Benchmark and terminal summaries are provided in Table 3 and Table 4.

Summary of 2025 Development Priorities

The 2025 assessment development documents identify three primary technical priorities:

1. Continued ADMB-to-RTMB bridging and validation.
2. Expanded sensitivity evaluation of data treatment and model assumptions.
3. Improved treatment of uncertainty associated with Russian removals and transboundary dynamics.

Introduction

This report provides an ASAR-formatted assessment narrative for EBS walleye pollock, consolidating material prepared for the 2025 CIE review cycle and subsequent September 2025 assessment development updates.

Stock ID

The assessed stock is EBS walleye pollock (*Gadus chalcogrammus*). The assessment supports AFSC scientific advice and NPFMC management decision-making, including annual ABC/OFL determinations.

Management Context

Recent assessment cycles have maintained continuity in the operational base model while expanding technical evaluations of uncertainty. In 2025, emphasis was placed on:

1. Demonstrating reproducible alignment between ADMB and RTMB formulations.
2. Evaluating sensitivity to data treatment (including survey influence and ageing methods).
3. Prioritizing unresolved uncertainty related to Russian removals and stock structure assumptions.

Fishery and Survey Context

The assessment integrates long-term fishery catch/composition information with fishery- independent survey indices, including BTS, ATS, and AVO-related data products. These information streams remain central to trend estimation, diagnostics, and sensitivity analysis.

Ecosystem Considerations

Ecosystem and climate context are included as supporting evidence (for example, condition and recruitment context, and multispecies mortality exploration), but are not yet implemented as direct ecosystem-coupled dynamic equations in the base operating model.

Data

Data treatment in this report follows the 2025 EBS pollock assessment workflow and preserves its primary fishery and survey information pathways.

Life History

Core life-history assumptions include maturity, growth, and natural mortality-at-age. The mean natural mortality represented in the standardized report object is 0.35.

Catch

Observed catch is imported from the ADMB base run through 2024. Terminal-year total catch in the standardized object is 1,300.

Indices and Standardization

Primary fishery-independent and auxiliary index streams are:

1. EBS bottom-trawl survey (BTS)
2. Acoustic-trawl survey (ATS)
3. AVO-related index information

Comparative index diagnostics retained in this report include Figure 18, Figure 14, and Figure 16.

Composition Data

Composition analyses in 2025 included direct contrast of traditional microscope ageing (TMA) and FT-NIR methods. Relevant diagnostics included in this report are:

- Cohort consistency: Figure 6 and Figure 5
- Proportion-at-age contrasts: Figure 37, Figure 36, Figure 35
- Input-structure schematic: Figure 8

Absolute Abundance

Absolute abundance is inferred through model-data integration of survey indices and composition information. No new direct absolute-abundance estimator is introduced in this ASAR pass.

Environmental and Ecosystem Indicators

Environmental and ecosystem indicators are retained as contextual evidence in this report version. No additional covariate modeling was introduced during the ASAR conversion step.

Assessment

Current Modeling Approach

The base assessment is the operational ADMB pollock model configuration used in recent AFSC cycles. In parallel, 2025 technical work evaluated RTMB implementation and bridging performance to ensure continuity of key quantities and diagnostics.

This ASAR report preserves the base-model results while standardizing delivery of tables, figures, and core summary quantities.

Configuration of the Base Model

The standardized model-results object (`std_output_admb.rda`) is assembled from base-run ADMB products and supplies the report preamble quantities used throughout the document.

Core report range and terminal summaries are:

- Year range: 1964-2024
- Terminal spawning biomass: 3,389.48
- Terminal fishing mortality: 0.1449
- Reference fishing mortality: 0.499

Bridging and Standardization

Two forms of bridging are represented:

1. **Model-platform bridging**: ADMB-to-RTMB consistency checks from the 2025 source work.
2. **Reporting-framework bridging**: translation of existing model outputs into ASAR-native tables and figures (see Table 4 and Figure 1).

Table 1: Selected ADMB optimization diagnostics from `pm.par` header metadata.

```
# A tibble: 3 x 2
  metric               value
  <chr>                <dbl>
1 Number of parameters 1366
2 Objective function   7294.
3 Maximum gradient        0
```

Modeling Results

Parameter Estimates

Key base-model quantities are summarized in Table 3. Important values used in this report include:

- Steepness: 0.622
- Unfished recruitment proxy (`R0`): 23,694
- Mean natural mortality: 0.35

Time Series

Time-series trajectories for spawning biomass and recruitment across model runs are shown in Figure 1. Terminal-year cross-model values are summarized in Table 4.

Model Fits

Model-fit diagnostics imported from the 2025 source workflow are retained in the figures section, including composition and index fit summaries (for example Figure 32, Figure 4, and Figure 2).

Model Diagnostics

Sensitivity Analyses

Sensitivity analyses retained from the 2025 assessment workflow are summarized below.

Data-Component Sensitivity

Leave-one-out evaluations indicate that model behavior is most sensitive to bottom-trawl survey information relative to other index components. Key diagnostics include:

- Relative SSB trajectories: Figure 22
- Recruitment trajectories: Figure 20
- Survey-index fits: Figure 18, Figure 14, Figure 16

Ageing and Composition Treatment

Uncertainty associated with ageing methods remains an active focus. Comparisons between TMA and FT-NIR products are shown in Figure 6, Figure 5, Figure 37, Figure 36, and Figure 35.

Natural Mortality and Structural Assumptions

Alternative mortality-related assumptions and associated effects are represented in retained sensitivity figures (for example Figure 31 and Figure 29).

Platform and Formulation Comparisons

The source workflow included comparisons across ADMB, RTMB, SS3, SAM, and related formulations to evaluate robustness of trend interpretation and identify implementation priorities.

Russian Removal Uncertainty

Review-response materials identify Russian removals as a high-priority unresolved uncertainty. That priority is carried forward here and remains central to upcoming scenario development.

Management Benchmarks

Reference-point quantities are drawn from the ADMB benchmark products and summarized in Table 3.

For the terminal year, key benchmark relationships are:

- `Fend` = 0.1449
- `Ftarg (reference F)` = 0.499
- `F/Ftarg` = 0.29
- `SBend` = 3,389.48
- `Bmsy proxy` = 2,400.99

These values are directly generated from the standardized report object and should be read in conjunction with the source 2025 benchmark interpretations.

Table 2: Near-term projection rows from `F40_t.rep` (years greater than terminal year).

```
# A tibble: 2 x 7
   Year   SSB `B/Bmsy` meanF  Fmsy `F/Fmsy`   F35
  <int> <dbl>    <dbl> <dbl> <dbl>    <dbl> <dbl>
1  2025 5227.     2.14 0.199 0.580    0.343 0.513
2  2026 4968.     2.04 0.199 0.580    0.343 0.513
```

Projections

Near-term projection rows are available in the benchmark file and are extracted below for years greater than the terminal assessment year.

Projection interpretation should follow the same assumptions and caveats described in the source 2025 assessment workflow.

Discussion

The ASAR conversion preserves the technical conclusions from the 2025 EBS pollock work while providing a reproducible, modular reporting structure for subsequent updates.

Three themes dominate the current evidence base:

1. ADMB and RTMB formulations can be aligned under matched assumptions, supporting platform modernization without loss of continuity.
2. Model outcomes remain sensitive to data treatment choices, especially survey weighting and ageing-method assumptions.
3. Russian-removal uncertainty remains a major unresolved axis for near-term scenario analysis.

Recommended priorities for the next update cycle are:

1. Extend the base run with the newest survey and fishery updates.
2. Maintain focused sensitivity/profiling work on weighting and process-variance settings.
3. Expand scenario communication through integrated benchmark/projection tables.
4. Continue closure of open review recommendations from the 2025 cycle.

This report can therefore serve as an operational draft scaffold for rapid update rather than as a static one-off conversion artifact.

Acknowledgements

Primary source materials for this bridged report were prepared by the EBS pollock assessment team, including Jim Ianelli, Carey McGilliard, and Sophia Wassermann.

This document was produced using the R package asar (Schiano et al. 2025), which is free to use and publicly available on GitHub.

References

Schiano, S., Breitbart, S., and Saul, S. 2025. Asar: Build NOAA stock assessment report. Available from https://github.com/nmfs-ost/asar.

Tables

Table 3: Reference-point summary from ADMB output for year 2024.

Year	SSB	Bmsy	B/Bmsy	meanF	Fmsy	F/Fmsy	F35
2,024	3,389.48	2,400.99	1.412	0.1987	0.499	0.398	0.4406

Table 4: Terminal year (2024) SSB and recruitment by model.

Model	Recruits	SSB
CEA, FE	51,371.32	3,208.31
CEA, RE	44,946.26	3,138.32
Pollock_model	18,286.00	3,389.48
SAM	20,801.03	2,352.90
SS3 Constrained	22,022.46	3,687.64
SS3 Flexible	23,672.36	3,203.65
SS3 base	22,173.41	3,308.61
Spock	41,317.98	4,998.88

Figures

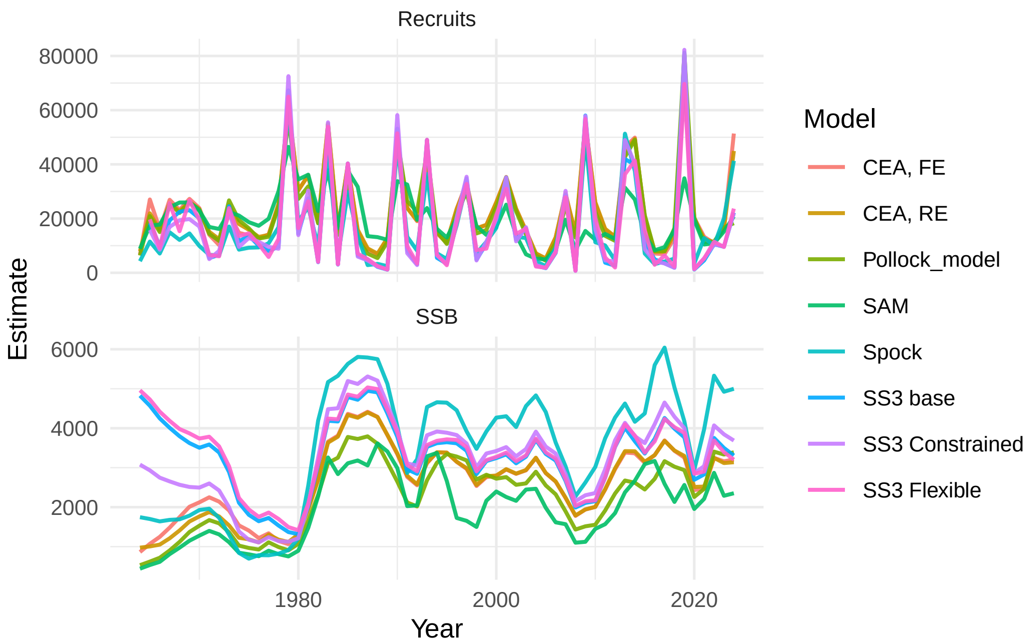

Figure 1: Model comparison of SSB and recruitment through 2024.

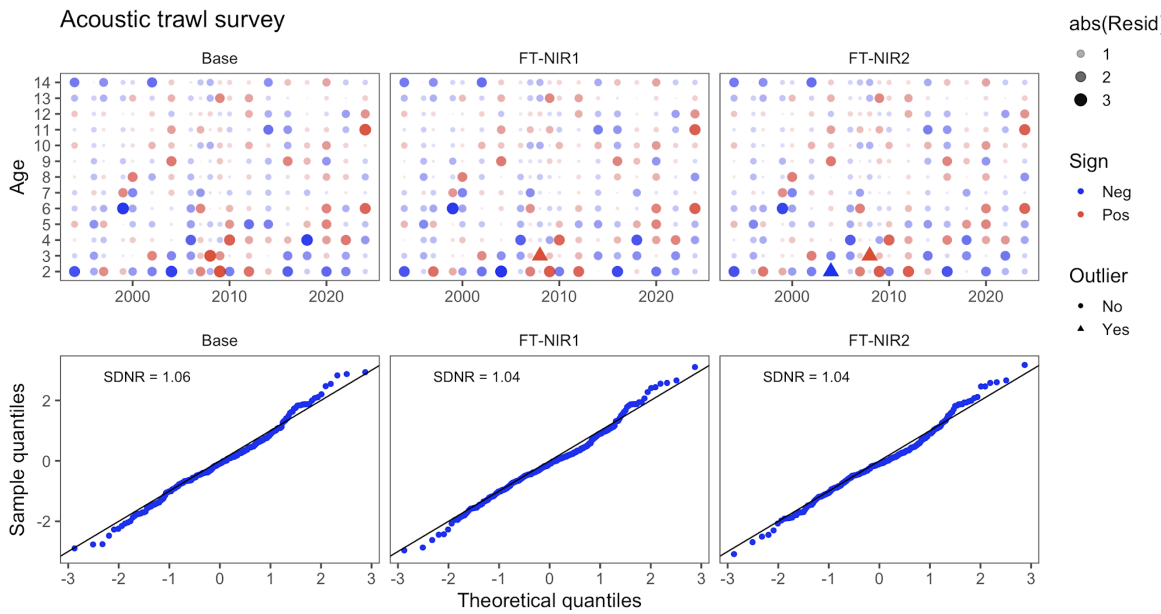

Figure 2: Ats Osa.

Appendices

Reproducibility Notes

This ASAR report is built from source files in:

- `/Users/jim/_mymods/noaa-afsc/ebs_pollock`
- `/Users/jim/_mymods/pollock/asar`

Key input files used in this build are listed below.

Figure and Table Packaging Notes

ASAR uses packaged `.rda` objects for programmatic inclusion:

1. Tables require `table` and `cap` fields (caption text may also be stored as `caption`).
2. Figures require `figure`, `cap`, and (for accessibility) `alt_text`.
3. Existing static figures can be included directly as PNG/JPG with explicit captions.

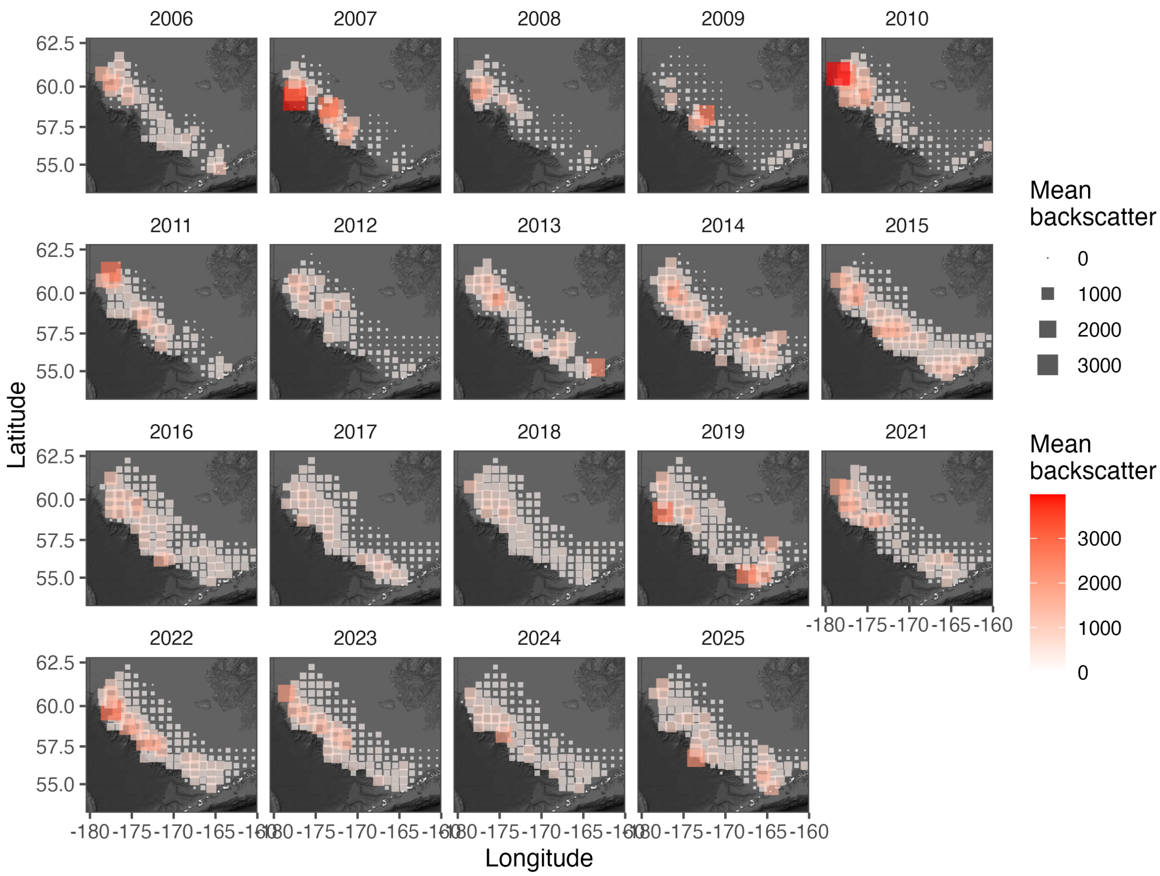

Figure 3: Avo Map Yr.

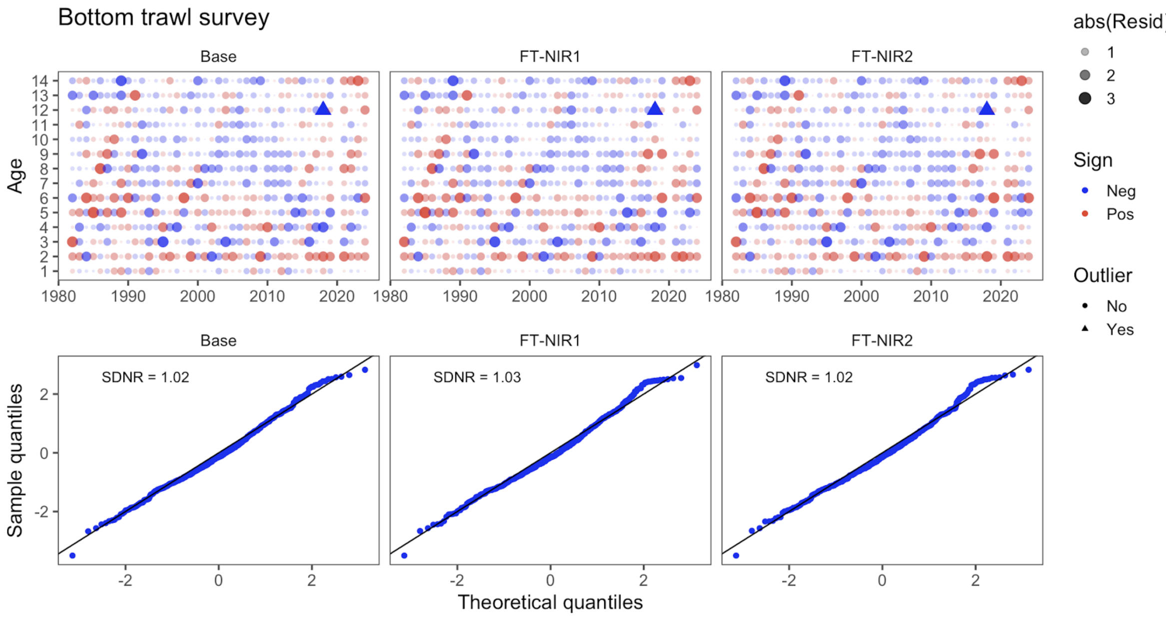

Figure 4: Bts Osa.

Table 1: Key source files used in this ASAR report build.

```
# A tibble: 5 x 3
  file                                                         exists modified_time
  <chr>                                                        <lgl>  <chr>
1 /Users/jim/_mymods/noaa-afsc/ebs_pollock/runs/lastyr/pm.~ TRUE   2025-09-
05 1~
2 /Users/jim/_mymods/noaa-afsc/ebs_pollock/runs/lastyr/pm.~ TRUE   2025-09-
05 1~
3 /Users/jim/_mymods/noaa-afsc/ebs_pollock/runs/lastyr/F40~ TRUE   2025-09-
05 1~
4 /Users/jim/_mymods/noaa-afsc/ebs_pollock/compares.qs      TRUE   2025-09-
02 1~
5 /Users/jim/_mymods/pollock/asar/report/std_output_admb.r~ TRUE   2026-02-
12 1~
```

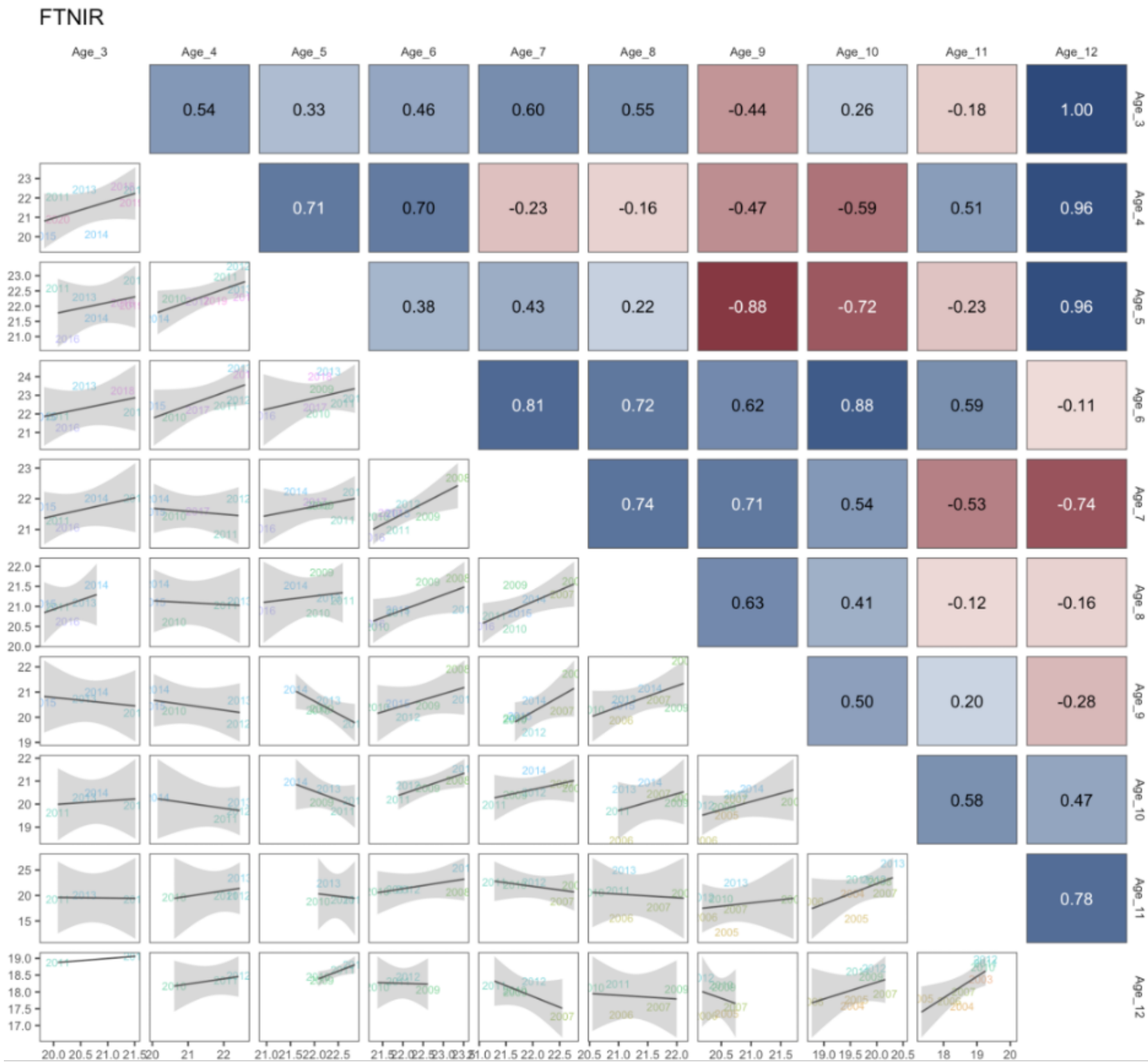

Figure 5: ConsisFTN.

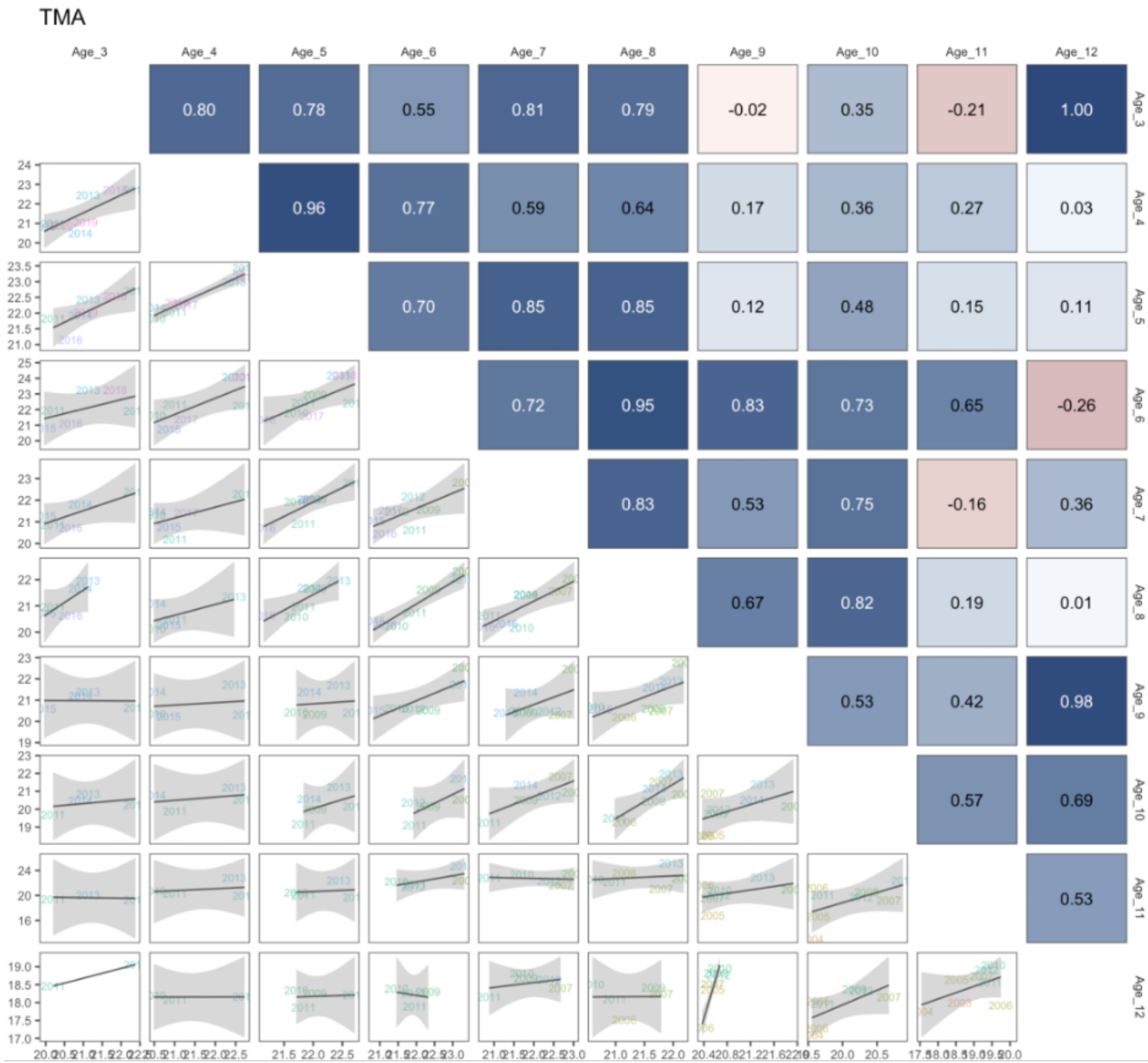

Figure 6: ConsisTMA.

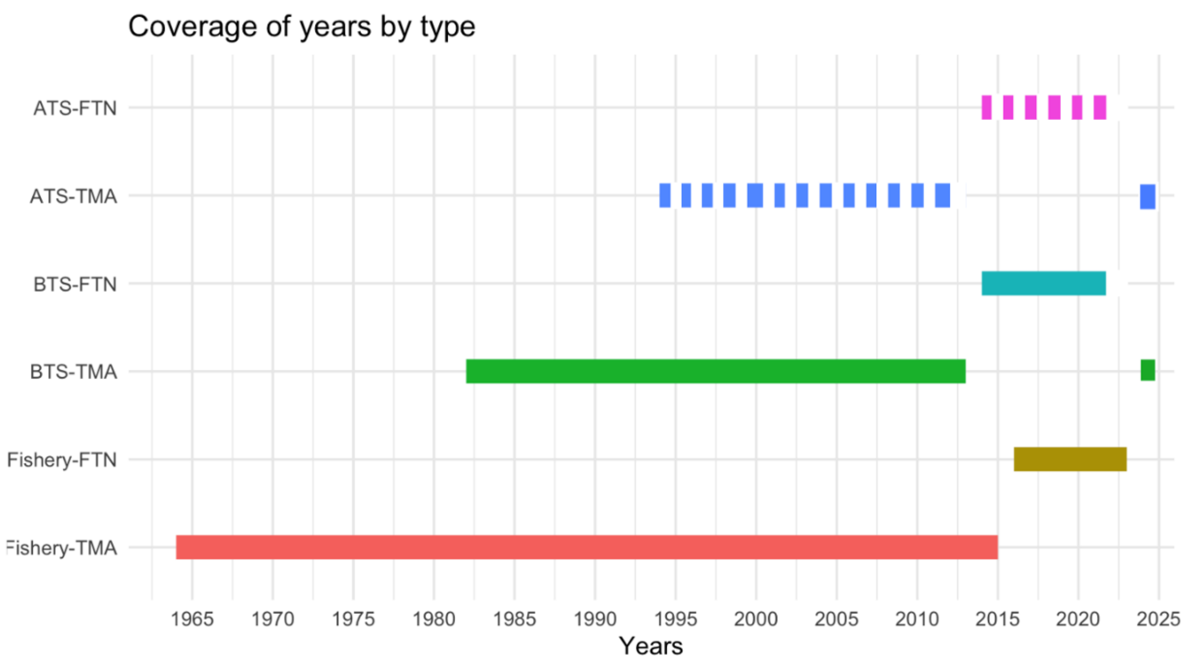

Figure 7: DataExtent.

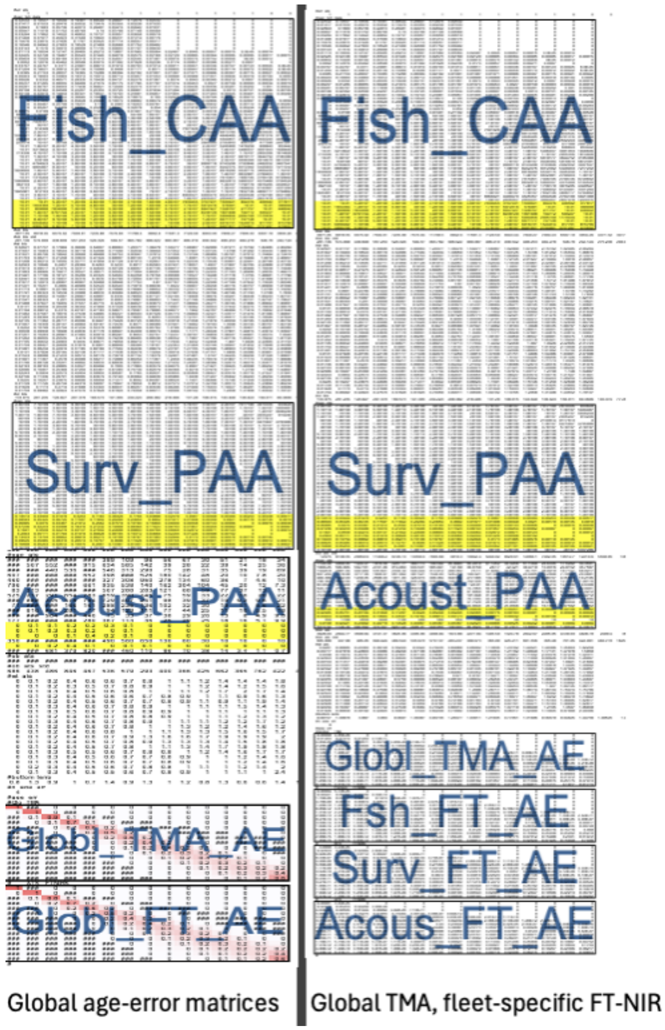

Figure 8: DataFile.

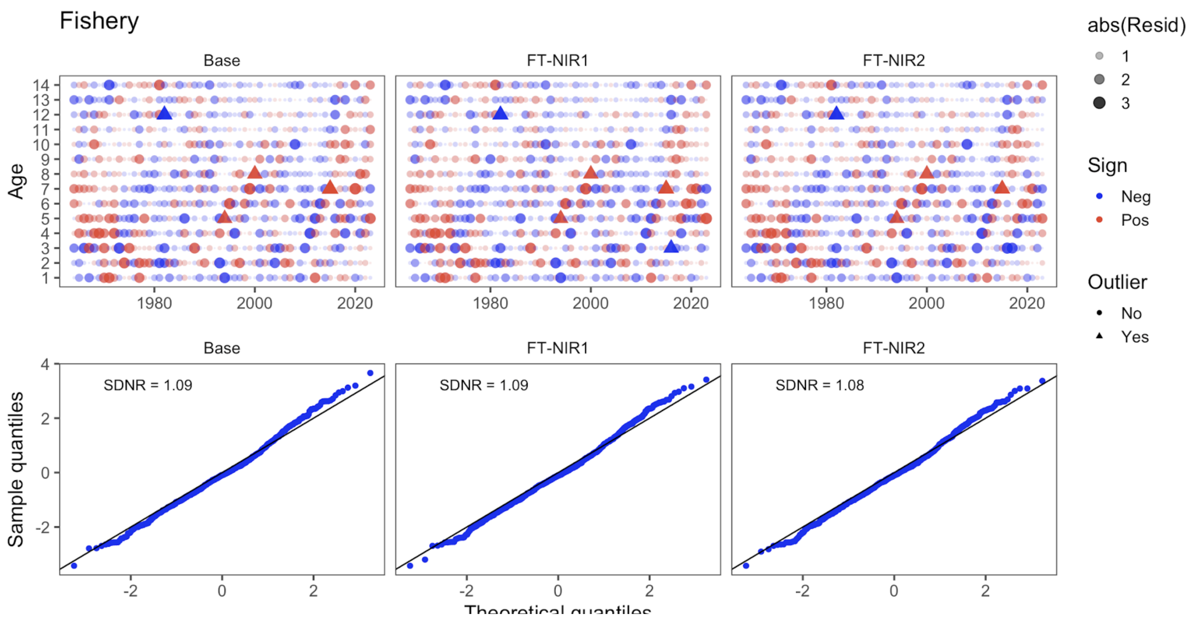

Figure 9: Fish Osa.

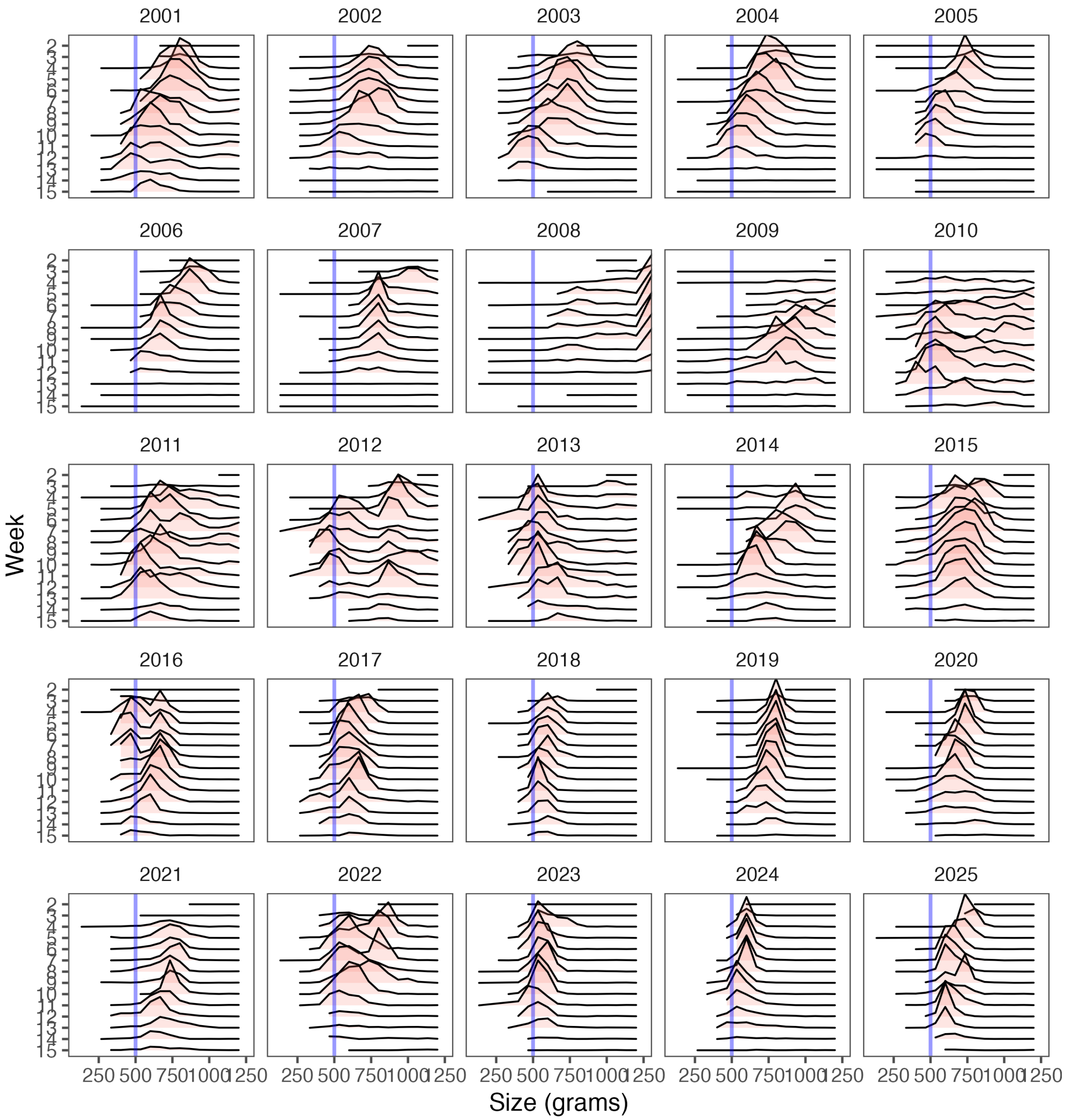

Figure 10: Fsh Wt Freq Week A.

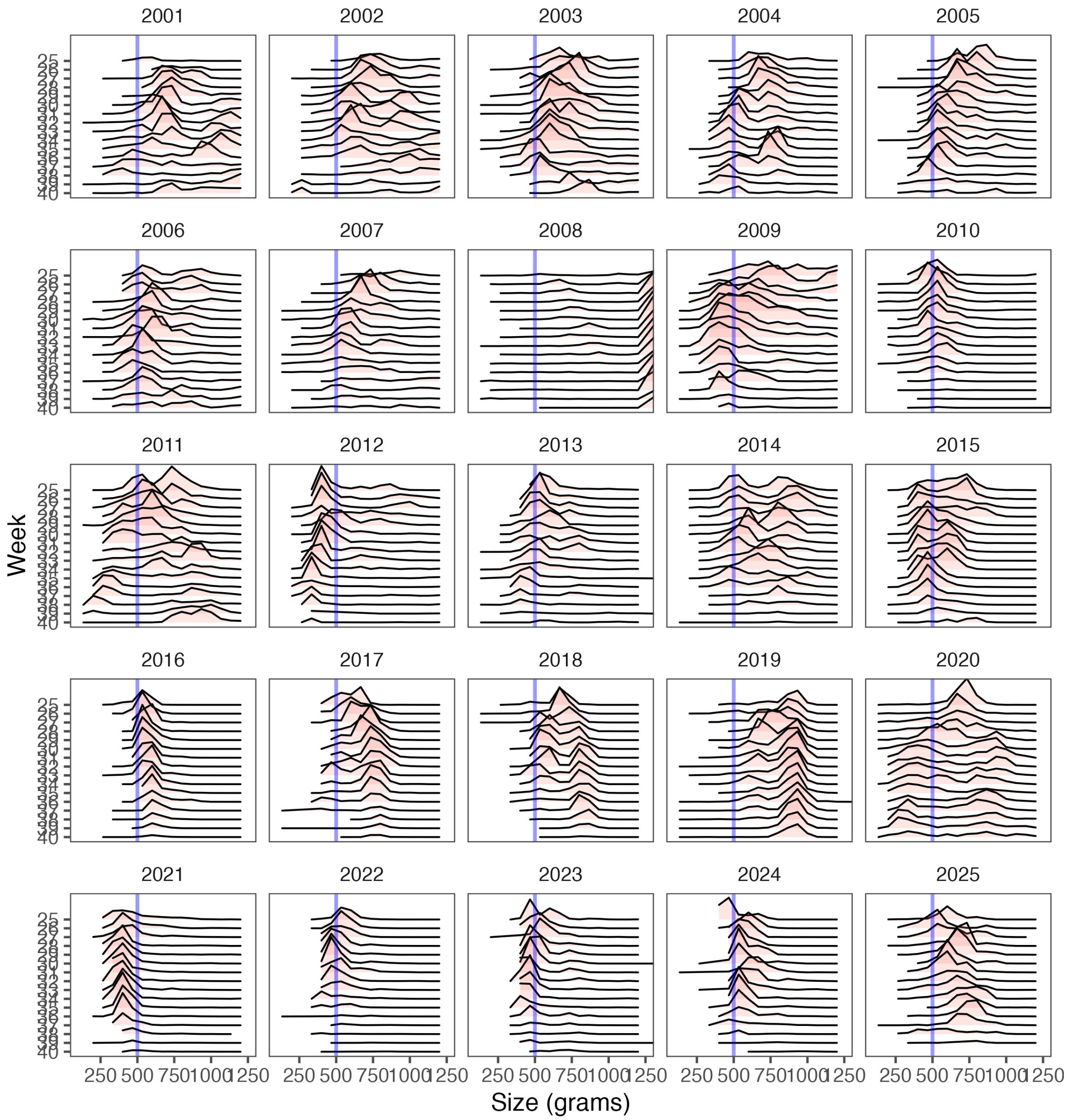

Figure 11: Fsh Wt Freq Week B.

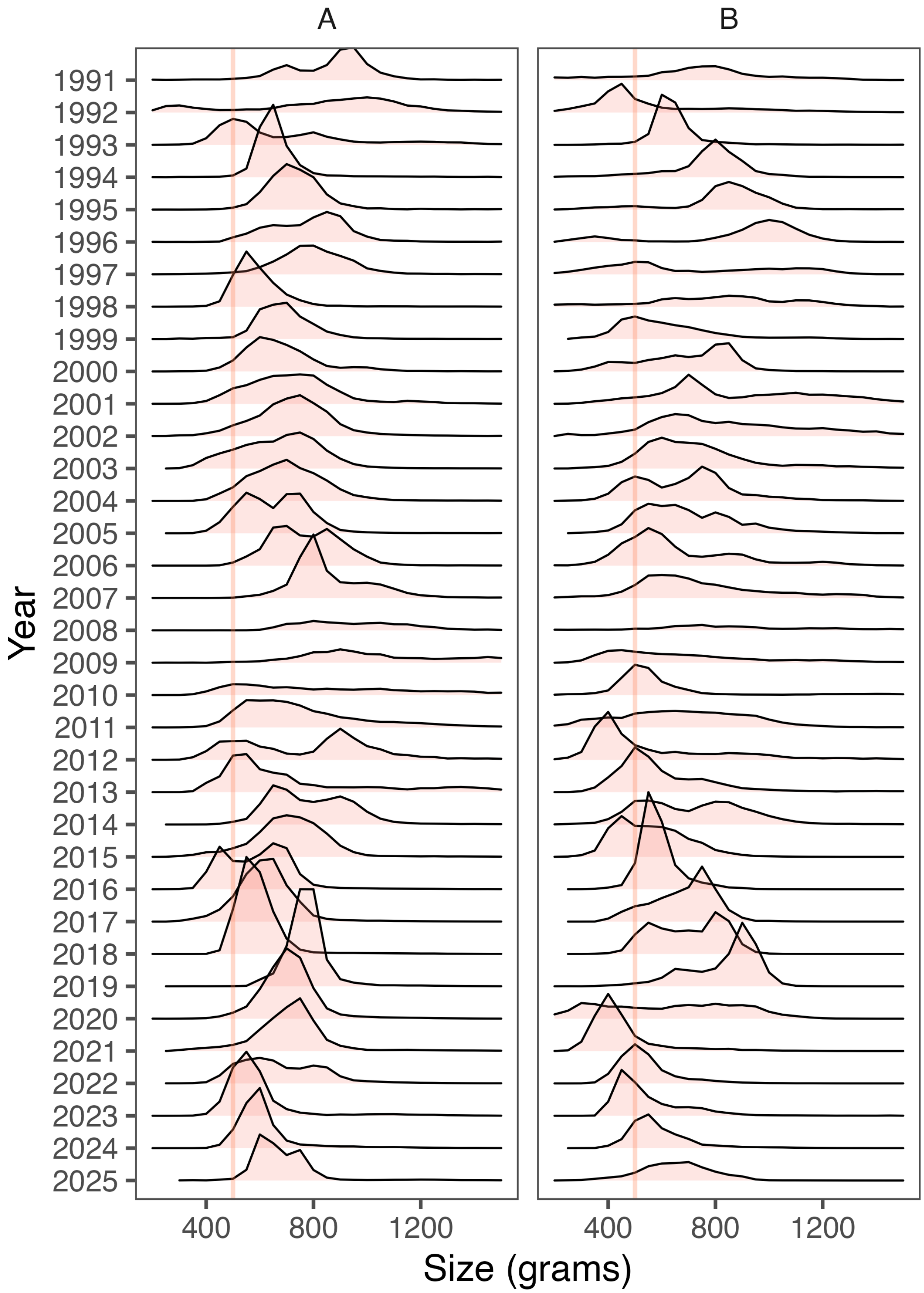

Figure 12: Fsh Wt Frq Yr.

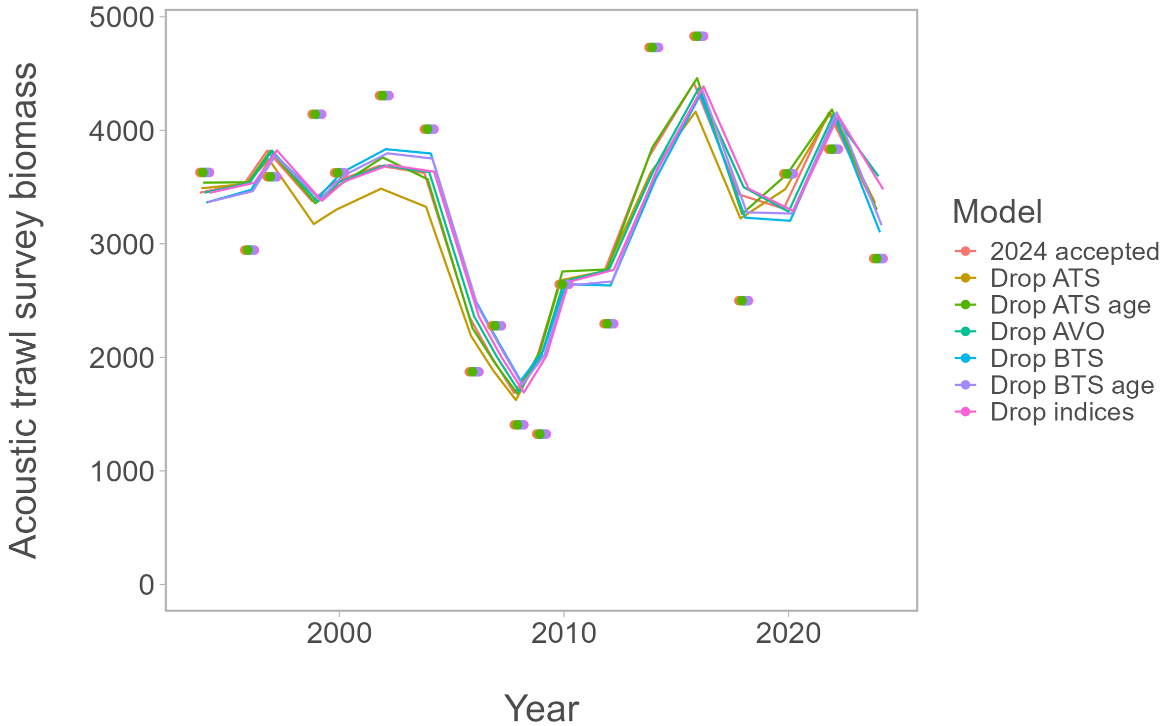

Figure 13: Loo Ats Index Present.

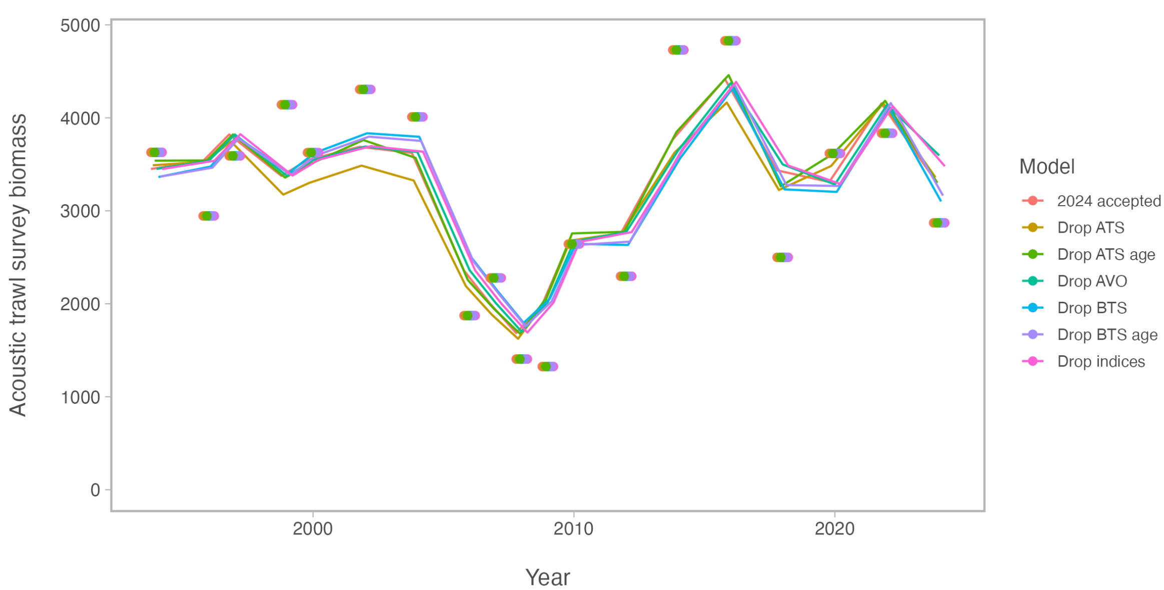

Figure 14: Loo Ats Index.

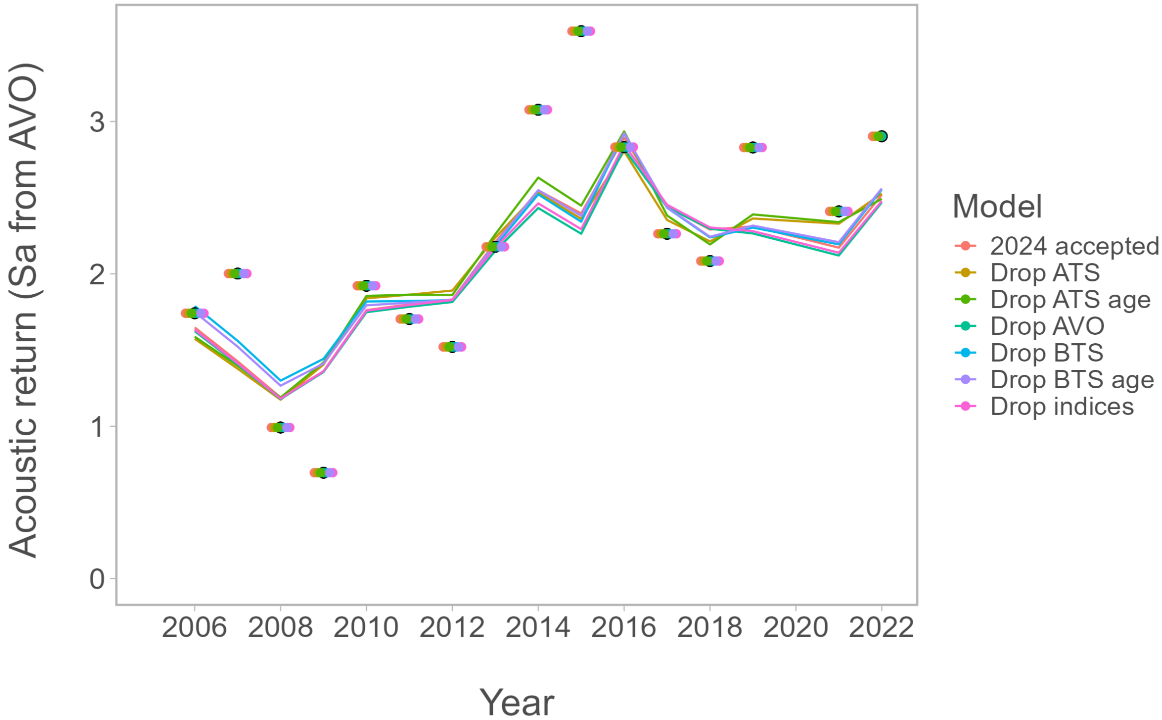

Figure 15: Loo Avo Index Present.

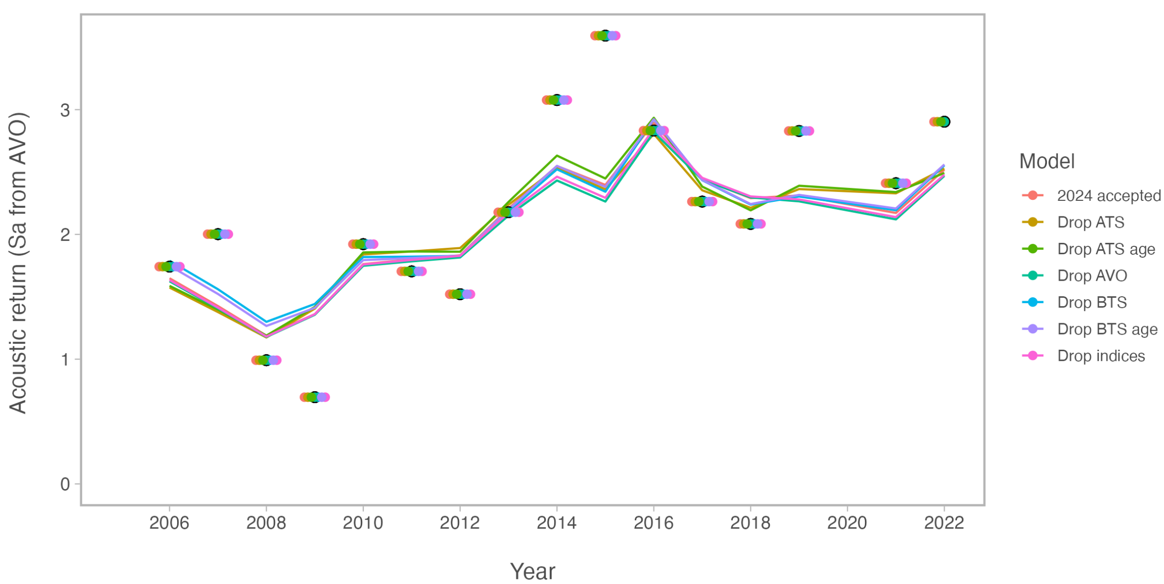

Figure 16: Loo Avo Index.

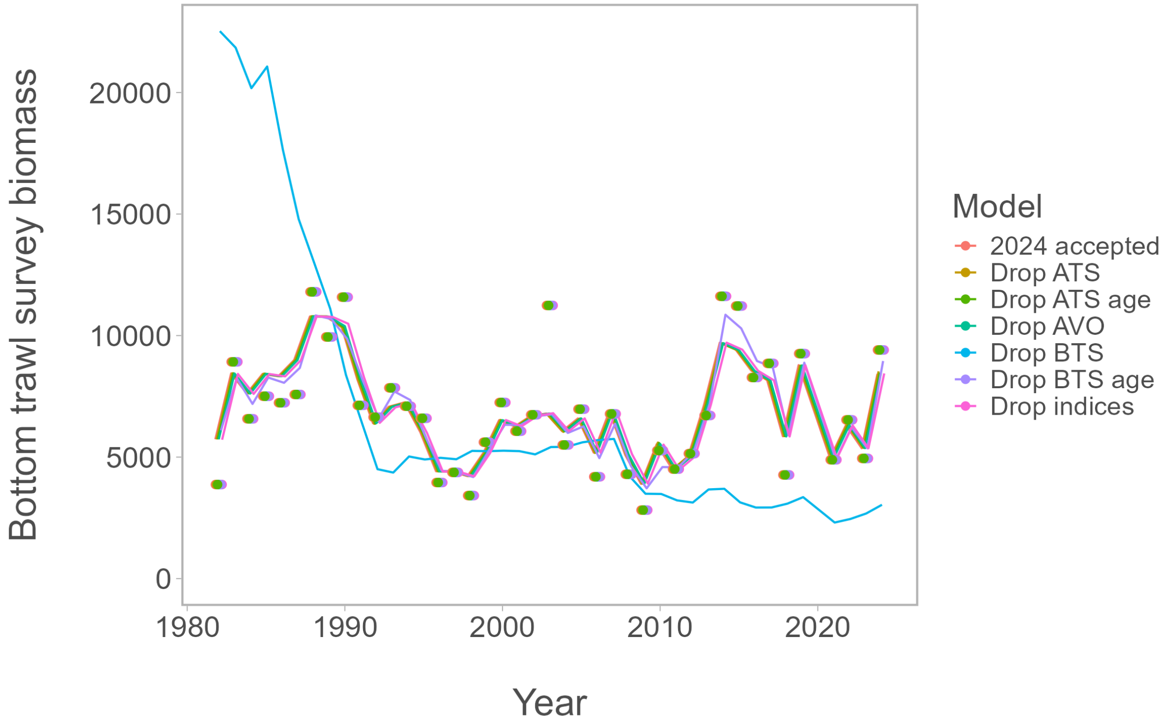

Figure 17: Loo Bts Index Present.

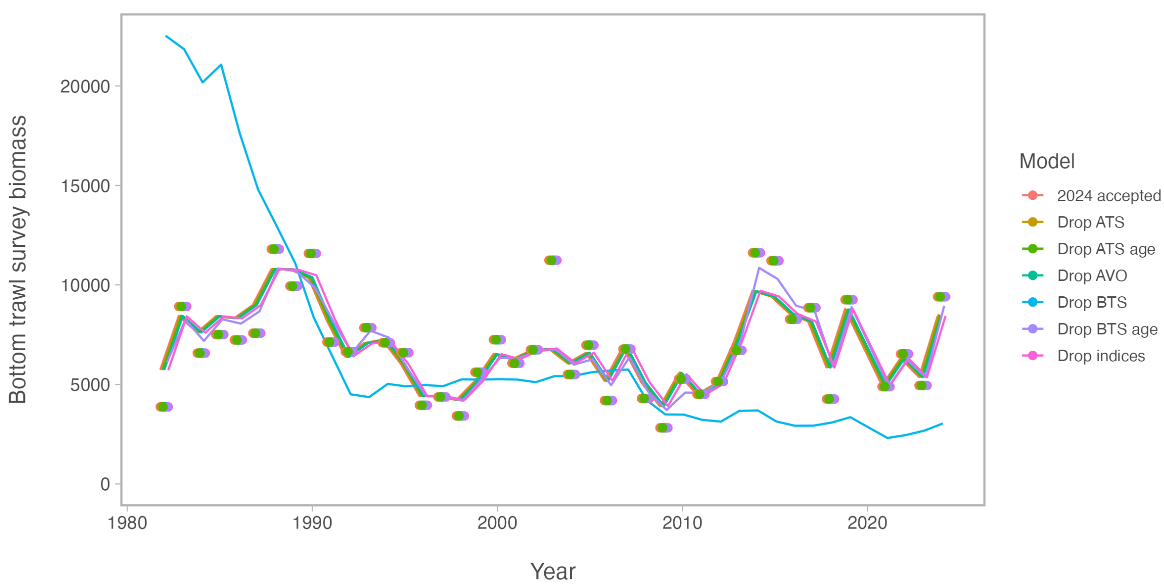

Figure 18: Loo Bts Index.

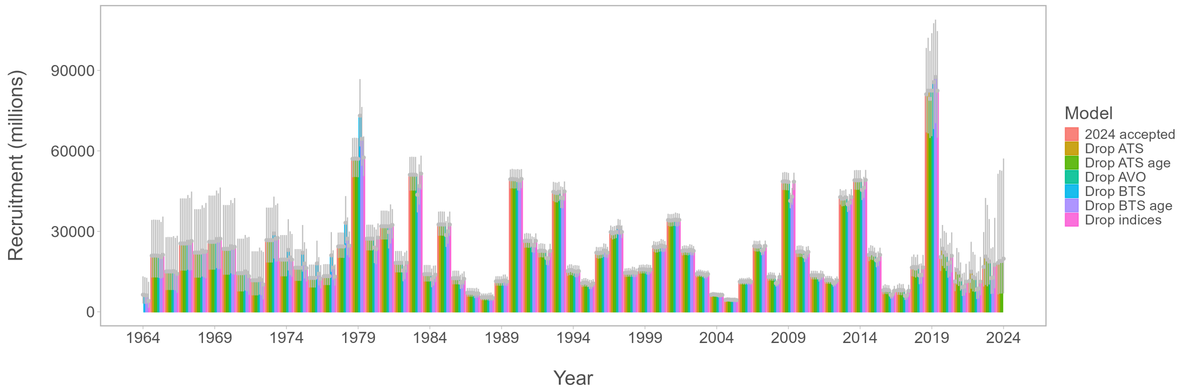

Figure 19: Loo Recruits Present.

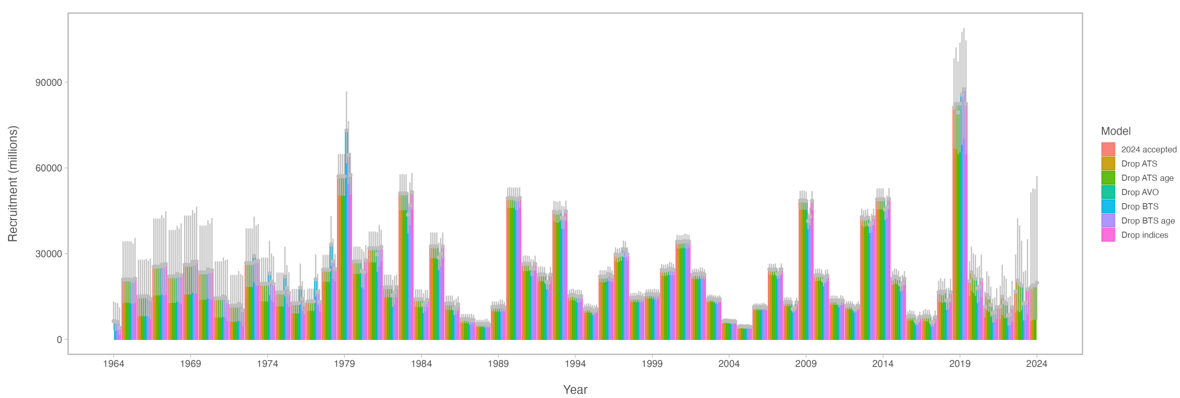

Figure 20: Loo Recruits.

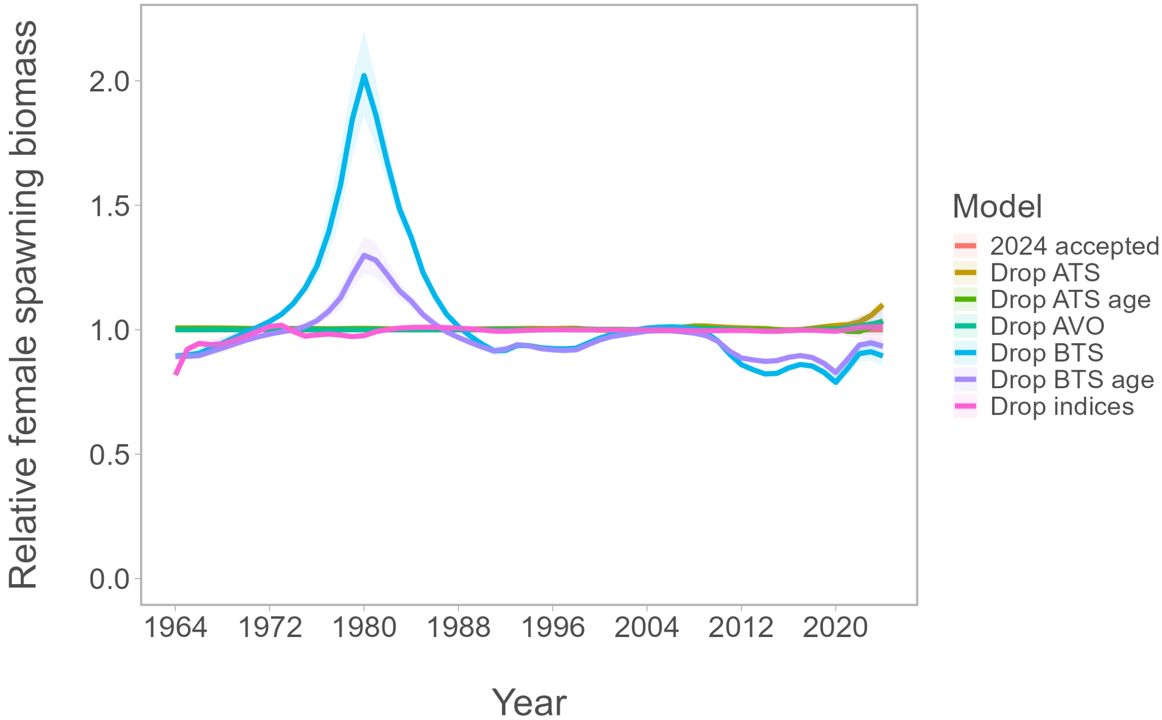

Figure 21: Loo Rel Ssb Present.

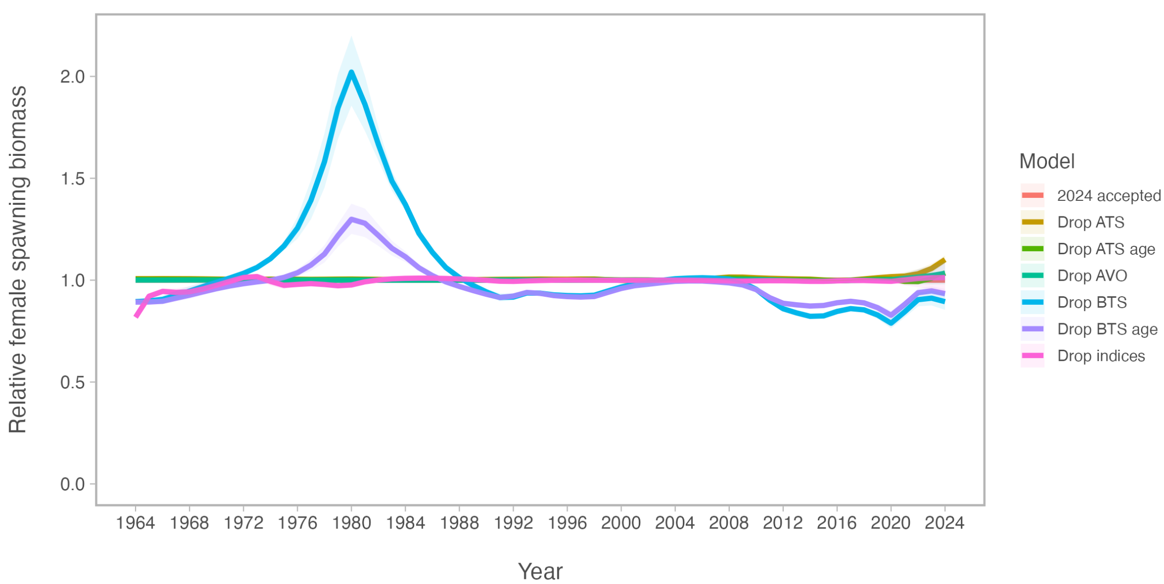

Figure 22: Loo Rel Ssb.

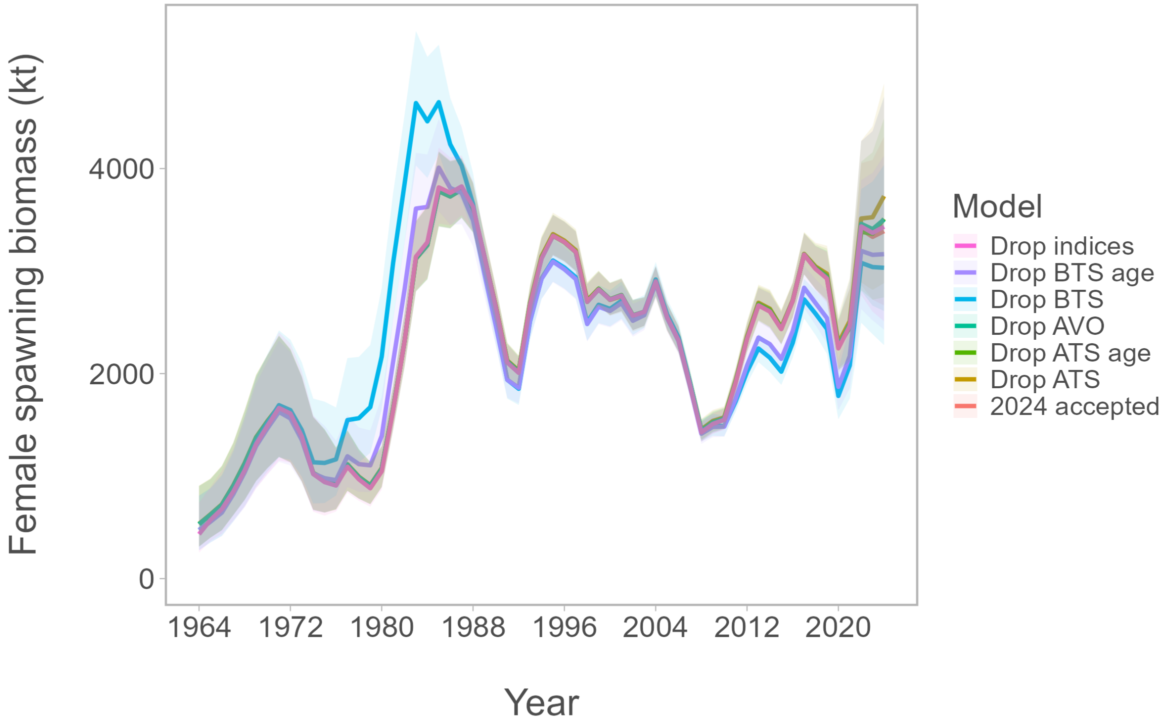

Figure 23: Loo Ssb Present.

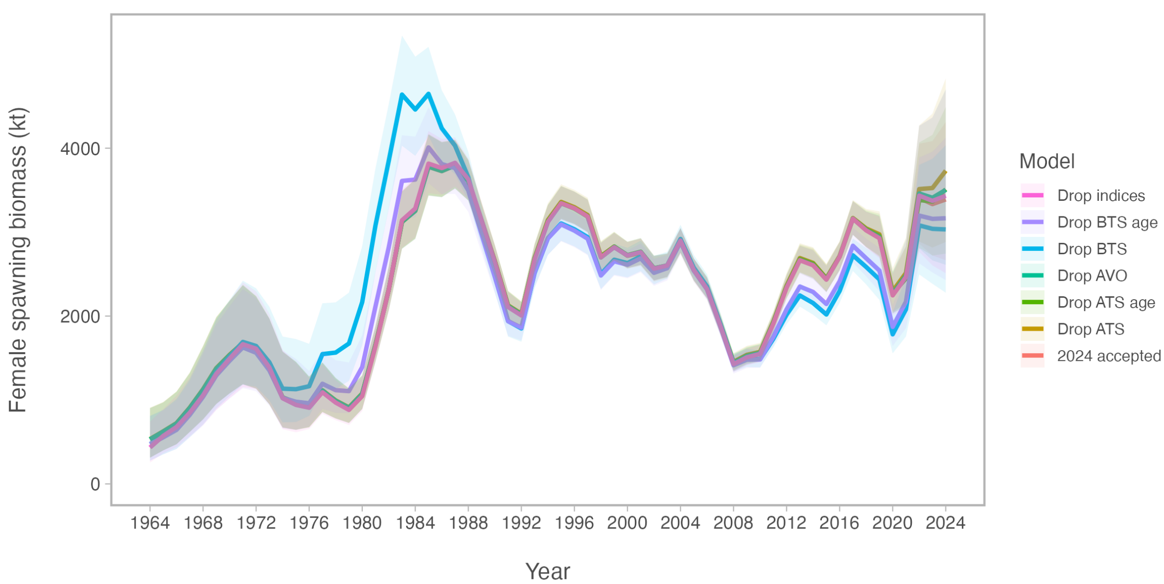

Figure 24: Loo Ssb.

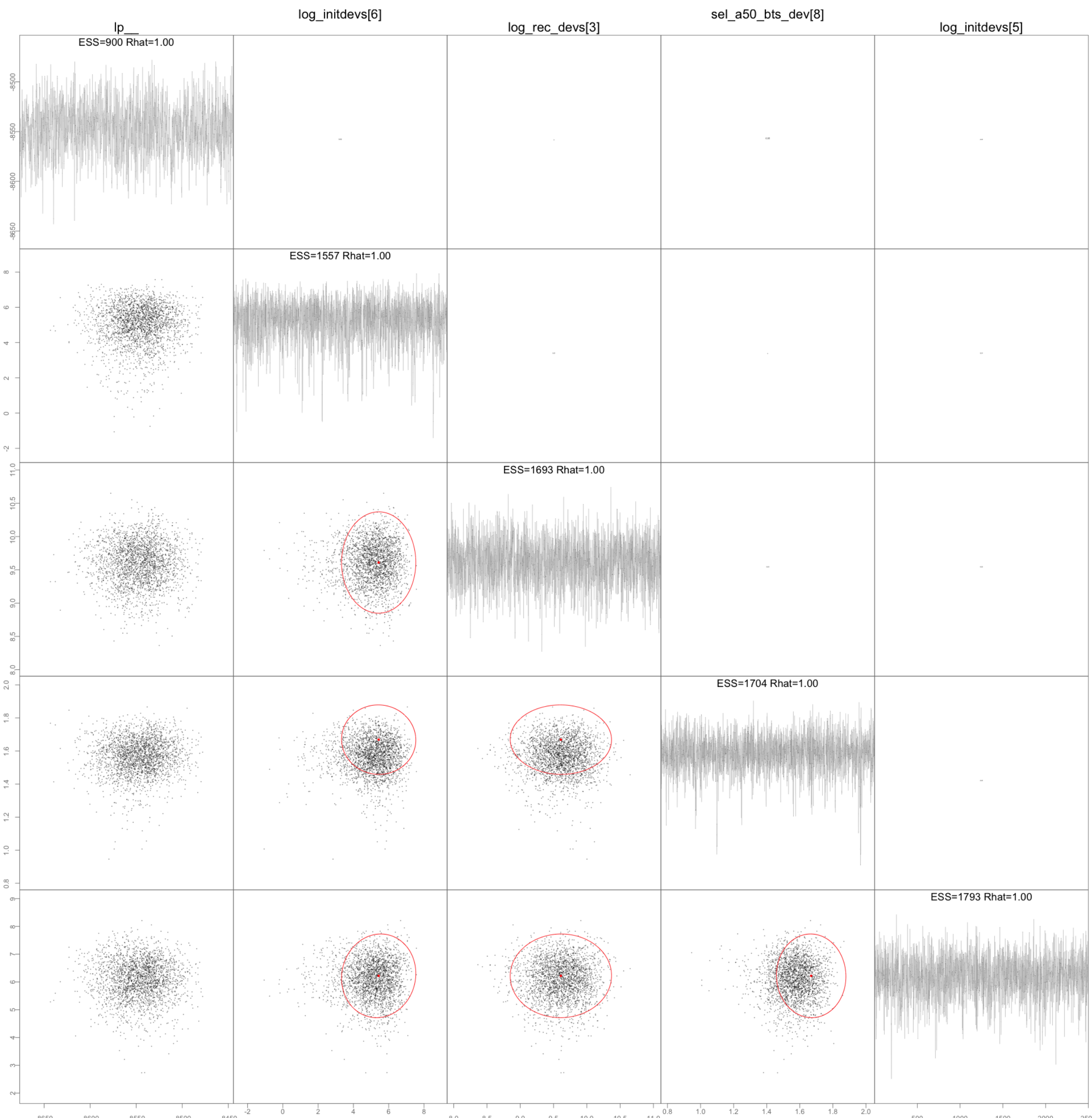

Figure 25: Mcmc Slow.

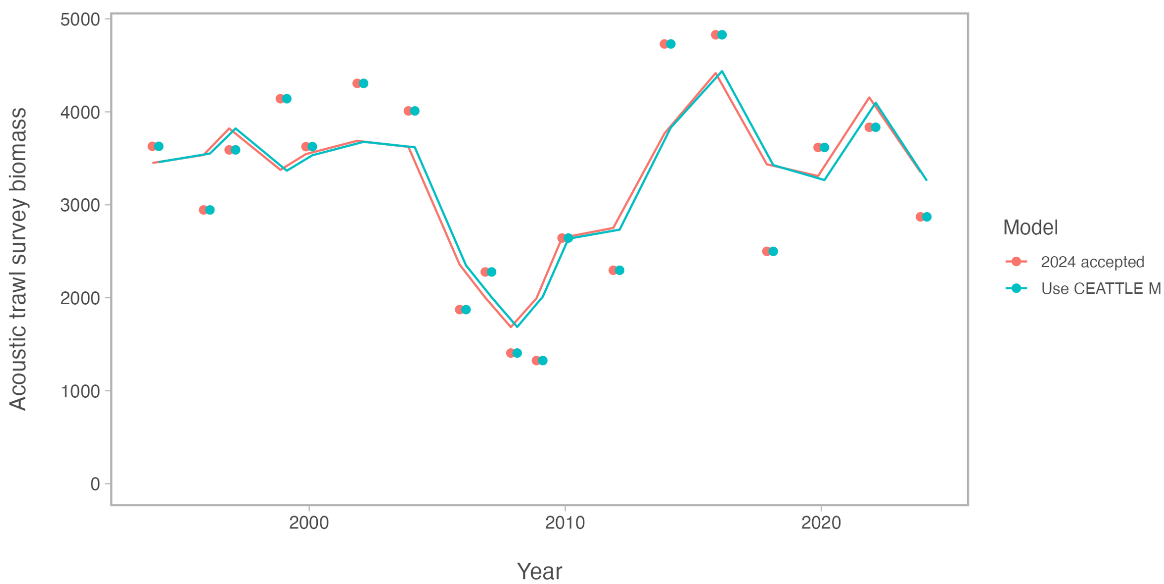

Figure 26: Mmat Ats Index.

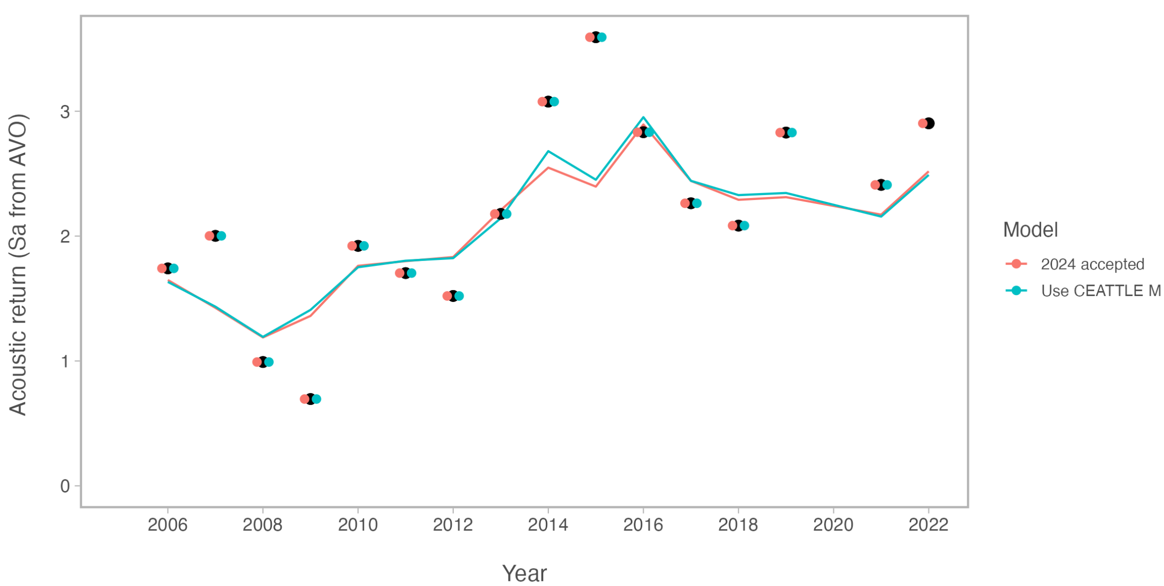

Figure 27: Mmat Avo Index.

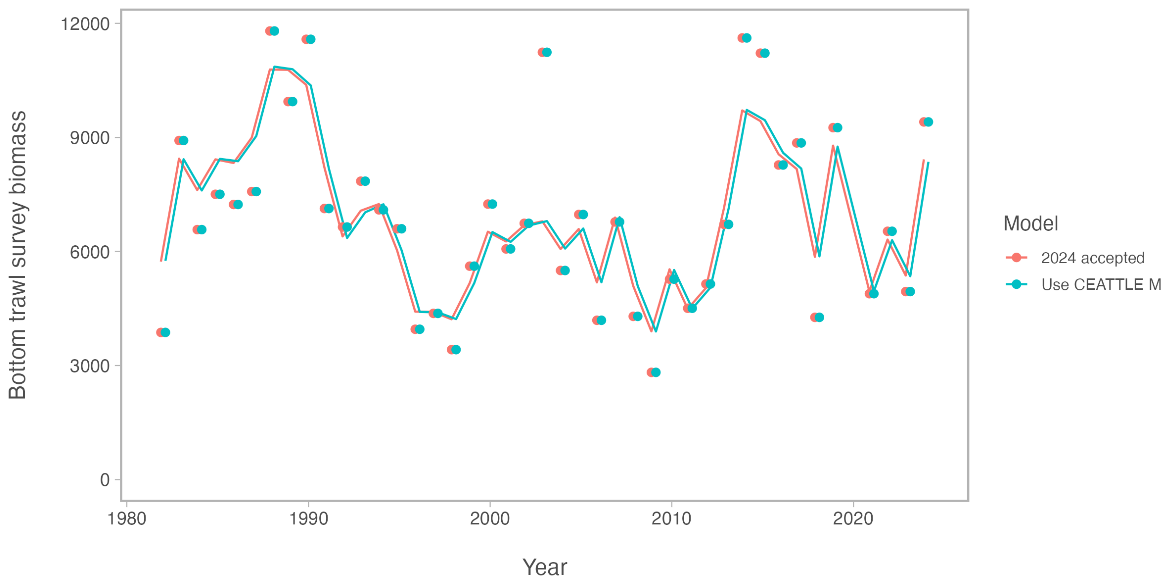

Figure 28: Mmat Bts Index.

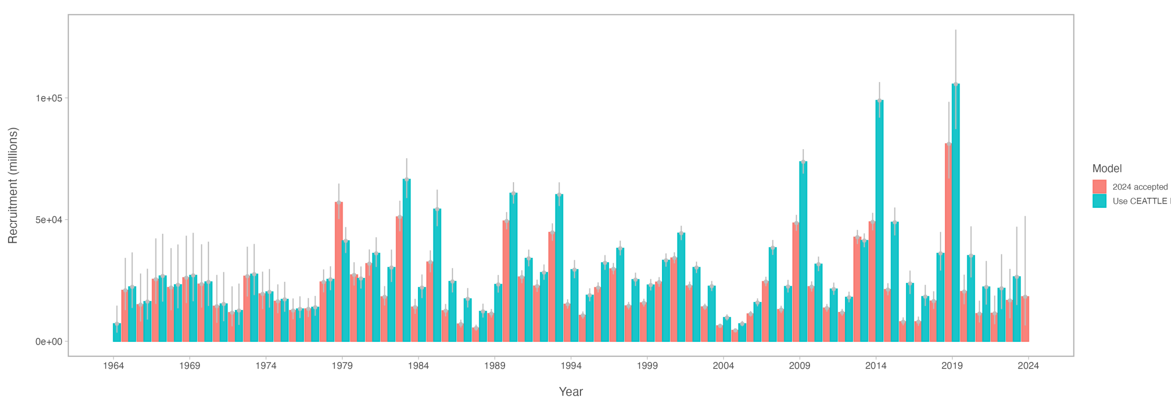

Figure 29: Mmat Recruits.

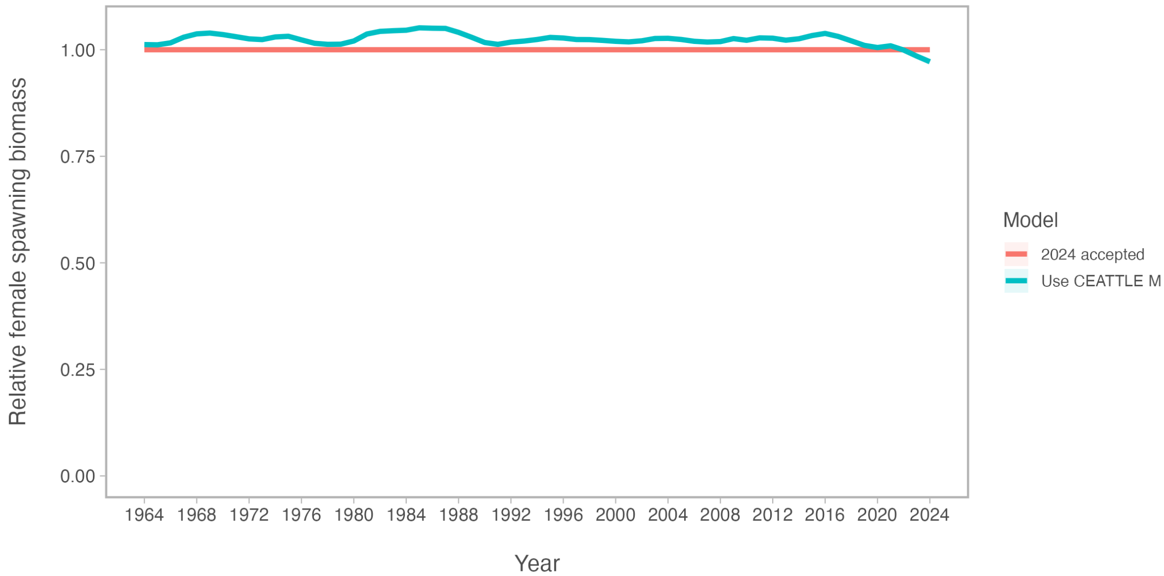

Figure 30: Mmat Rel Ssb.

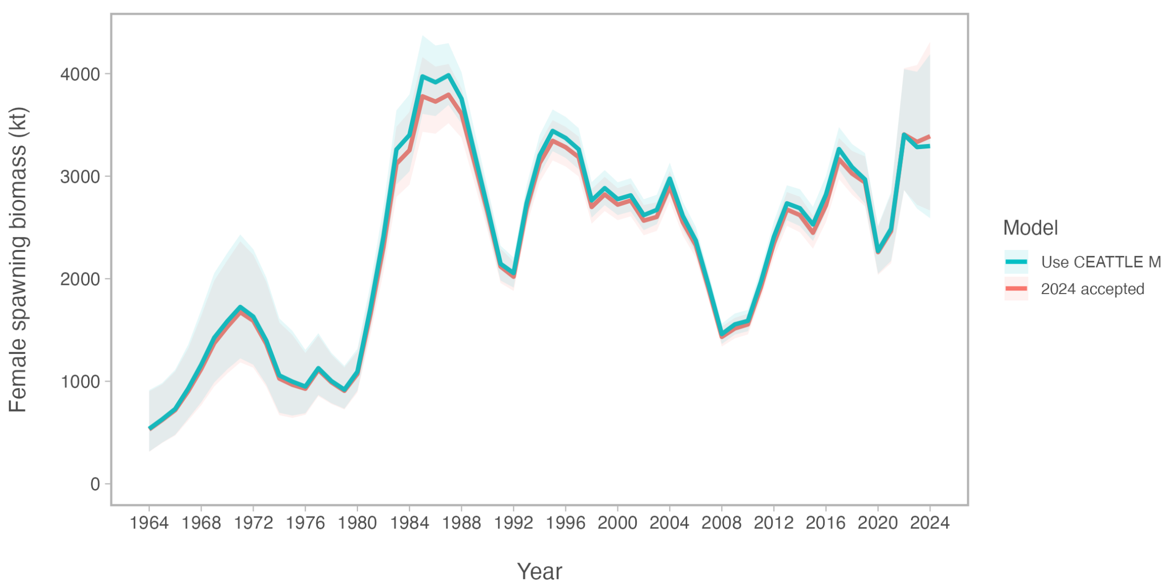

Figure 31: Mmat Ssb.

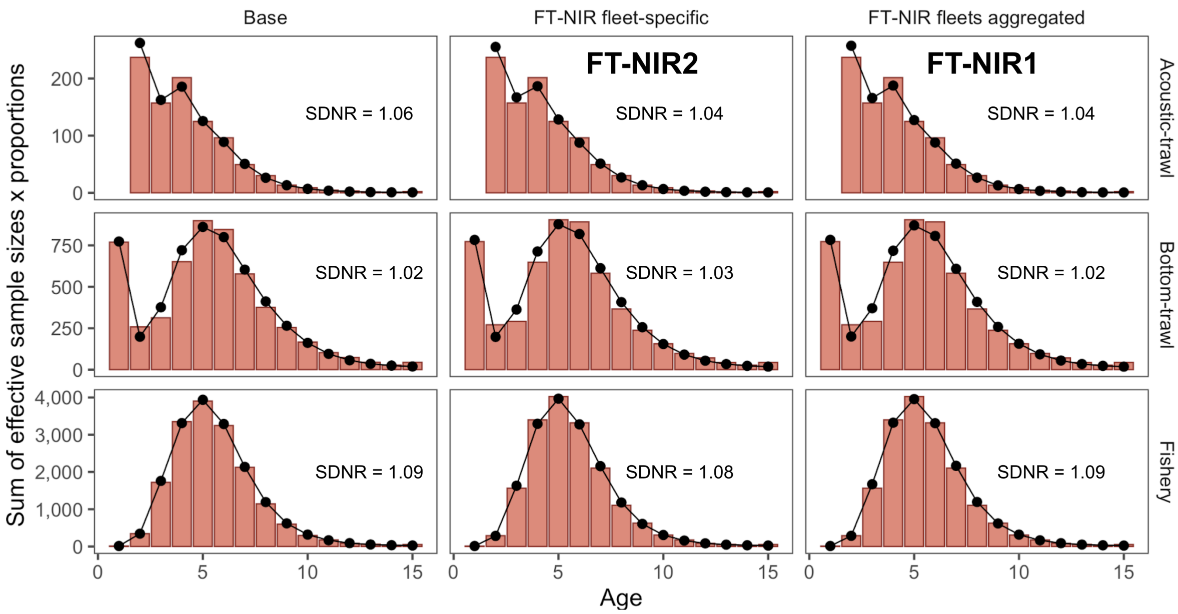

Figure 32: Mod Fit Agg Comp.

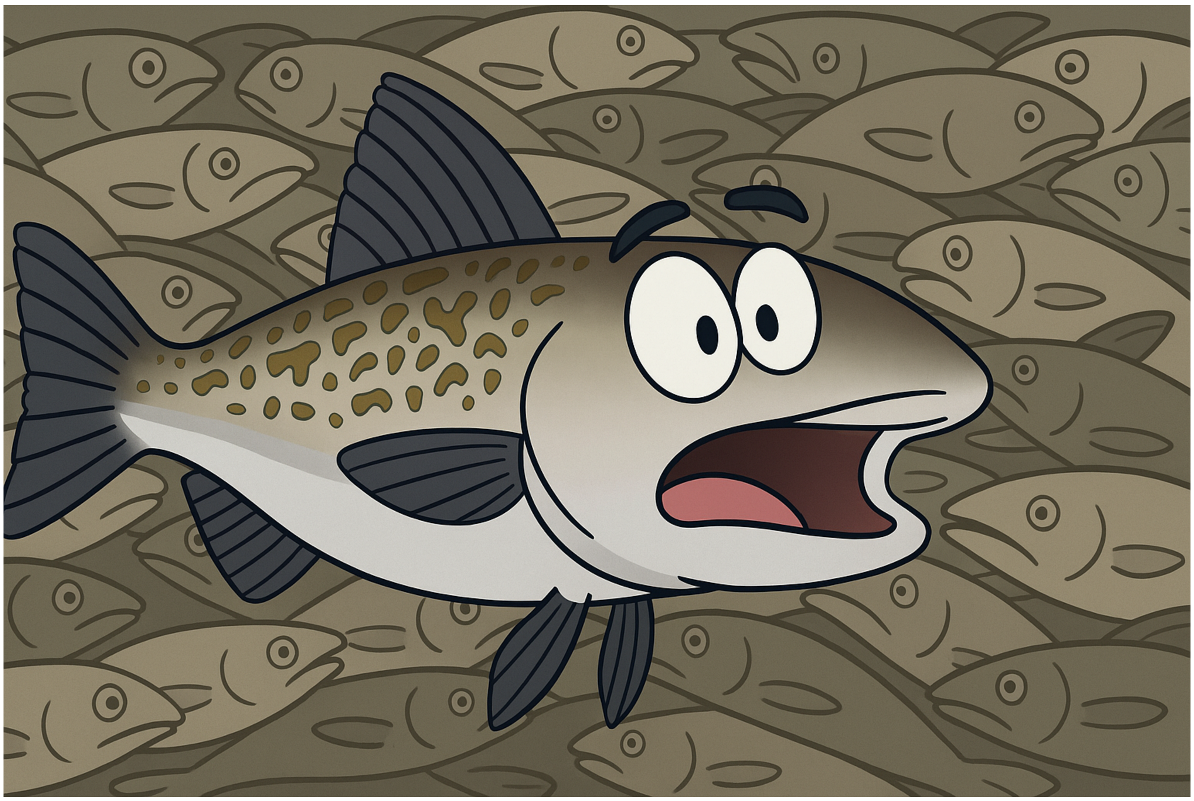

Figure 33: Pollock Cartoon.

Figure 34: Pollock.

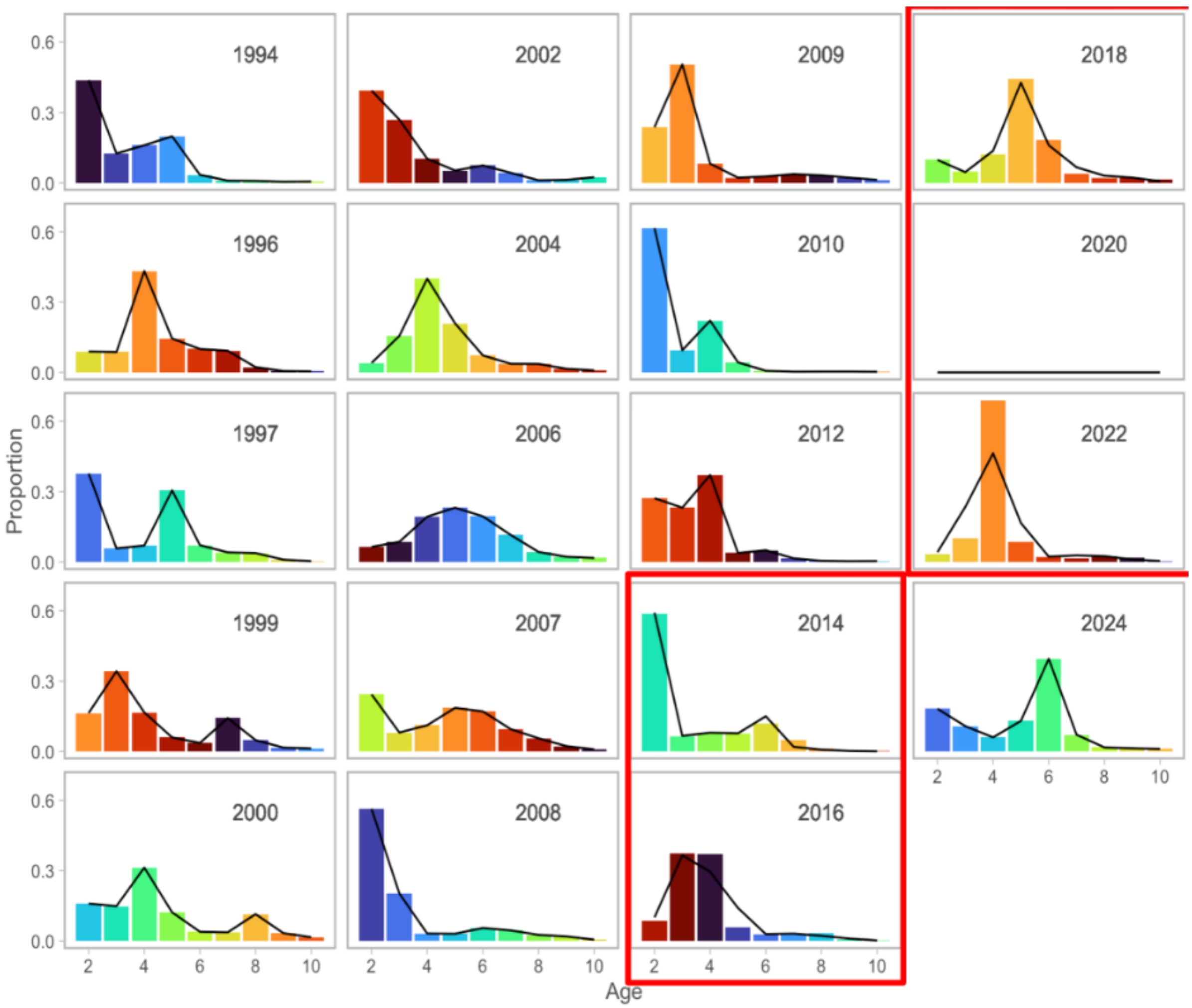

Figure 35: ProportionsATS.

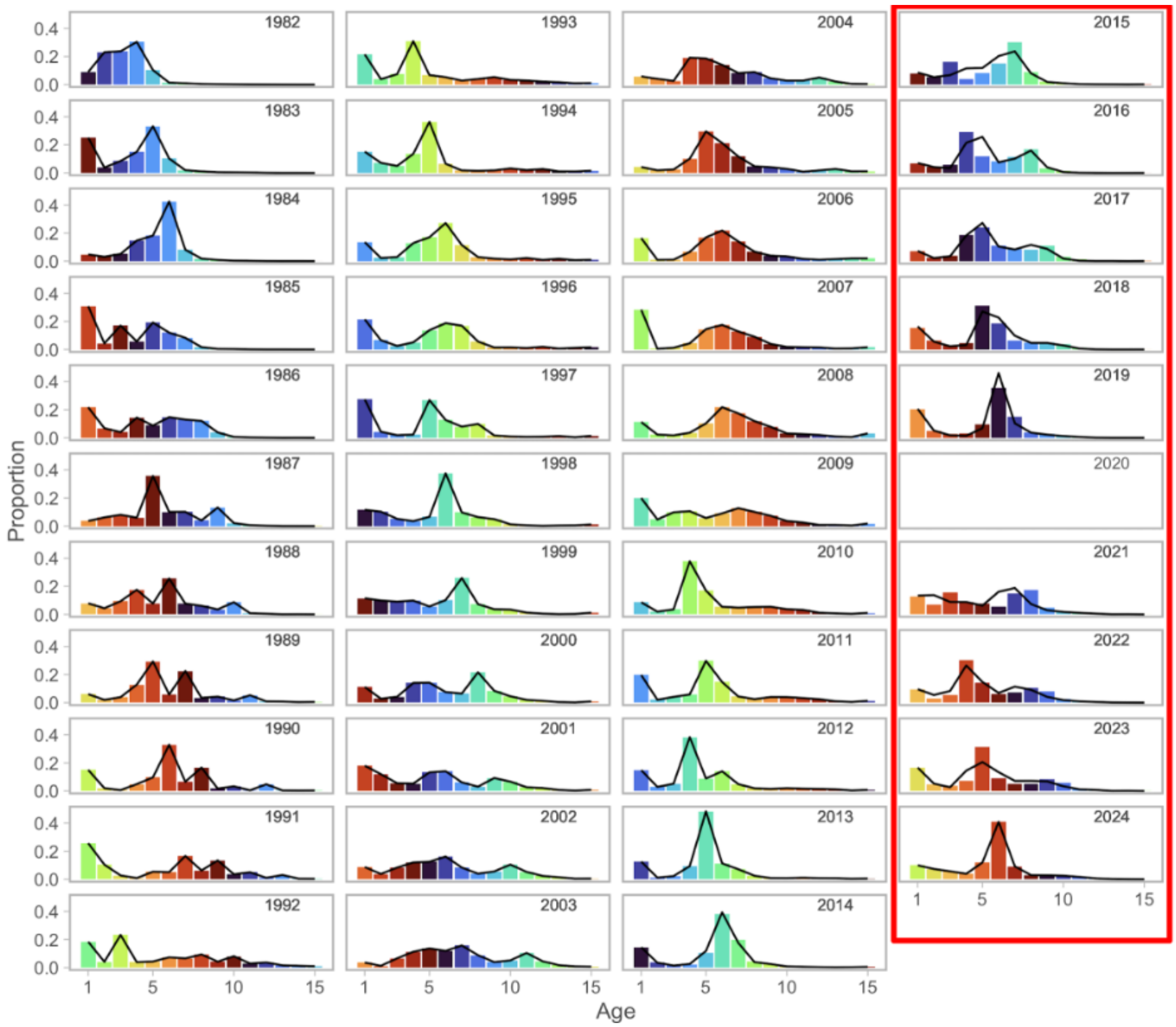

Figure 36: ProportionsBTS.

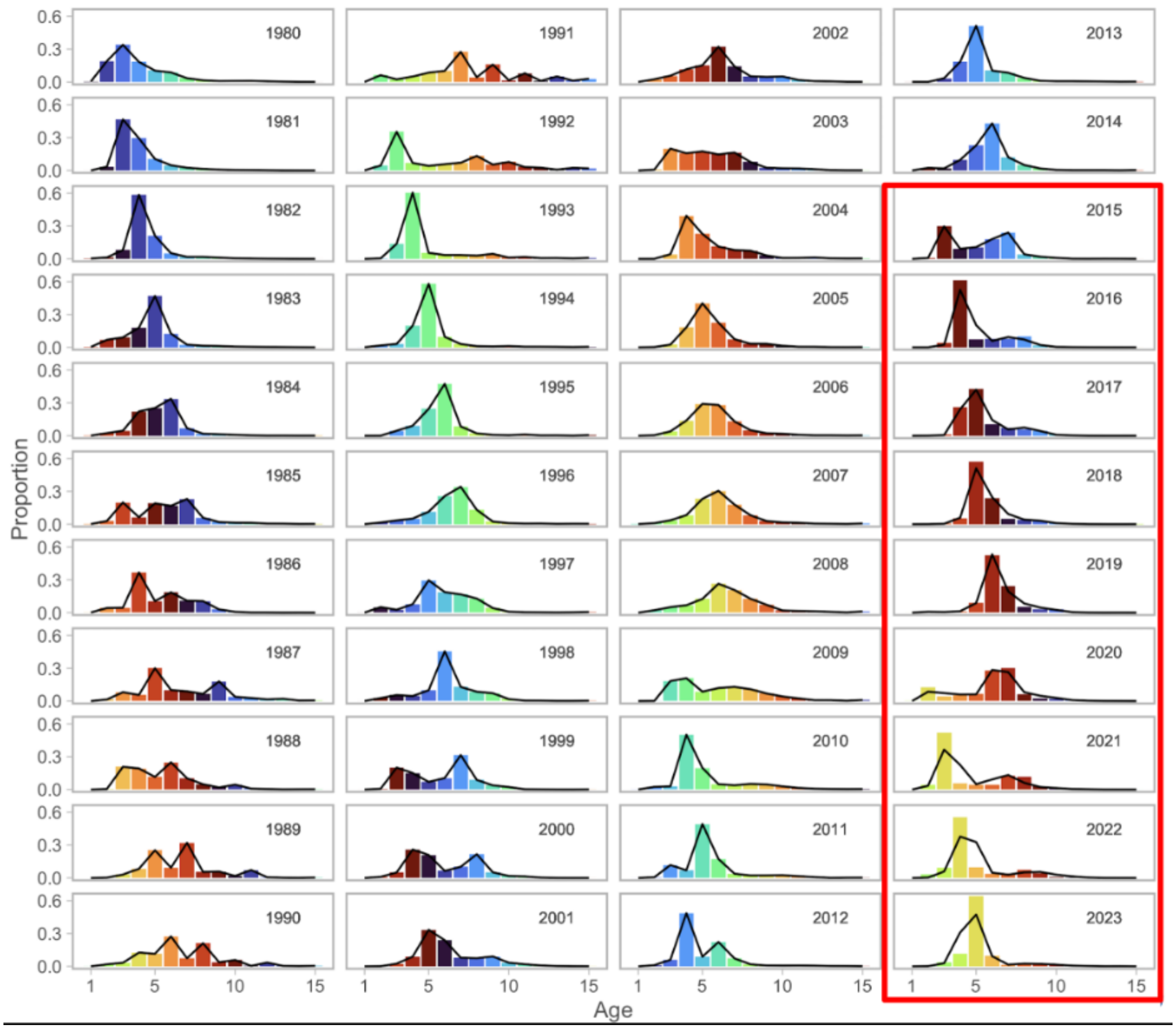

Figure 37: ProportionsFSH.

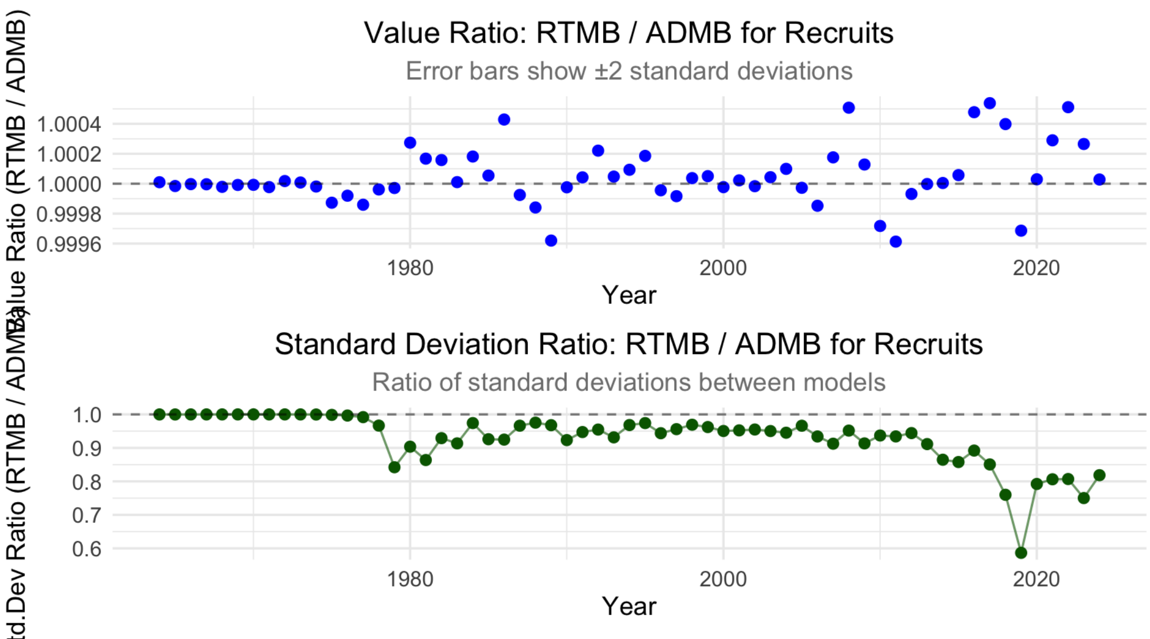

Figure 38: Rec Ratio.

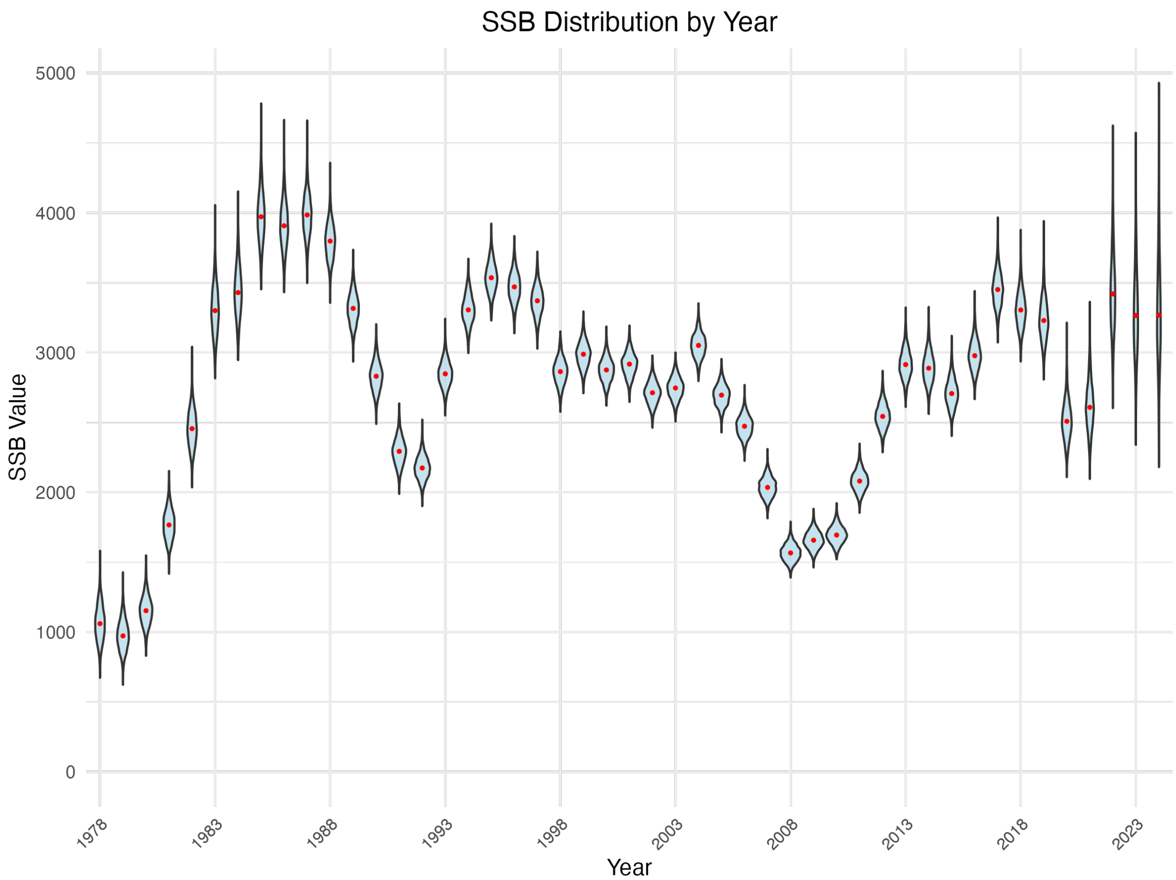

Figure 39: SSB Mcmc.

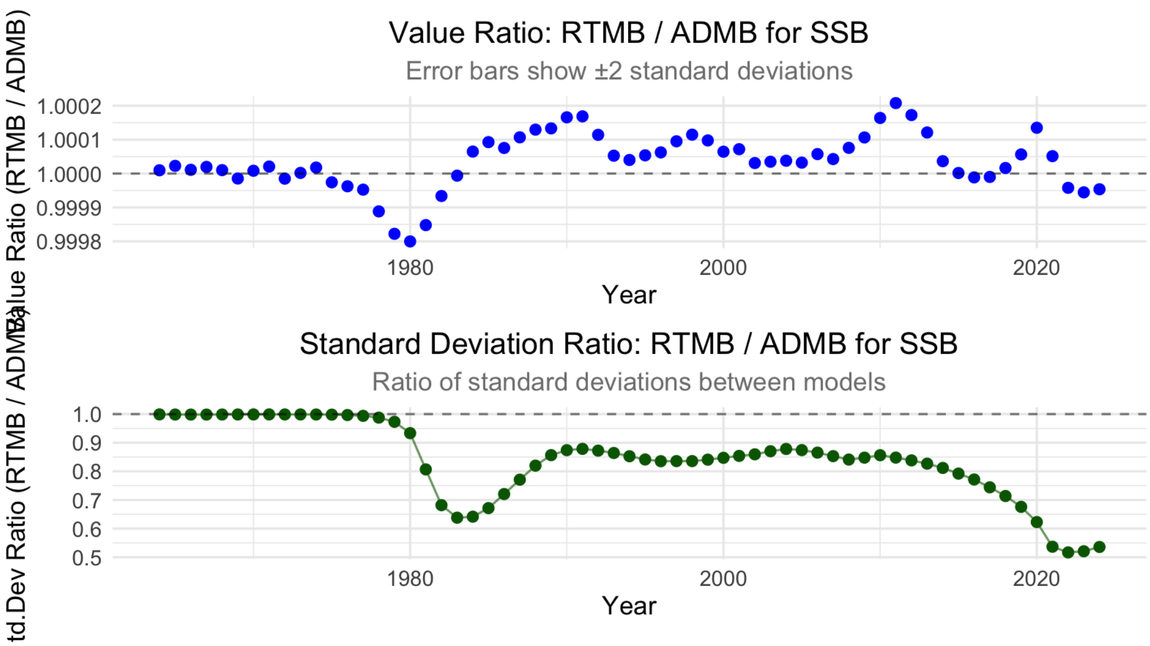

Figure 40: Ssb Ratio.

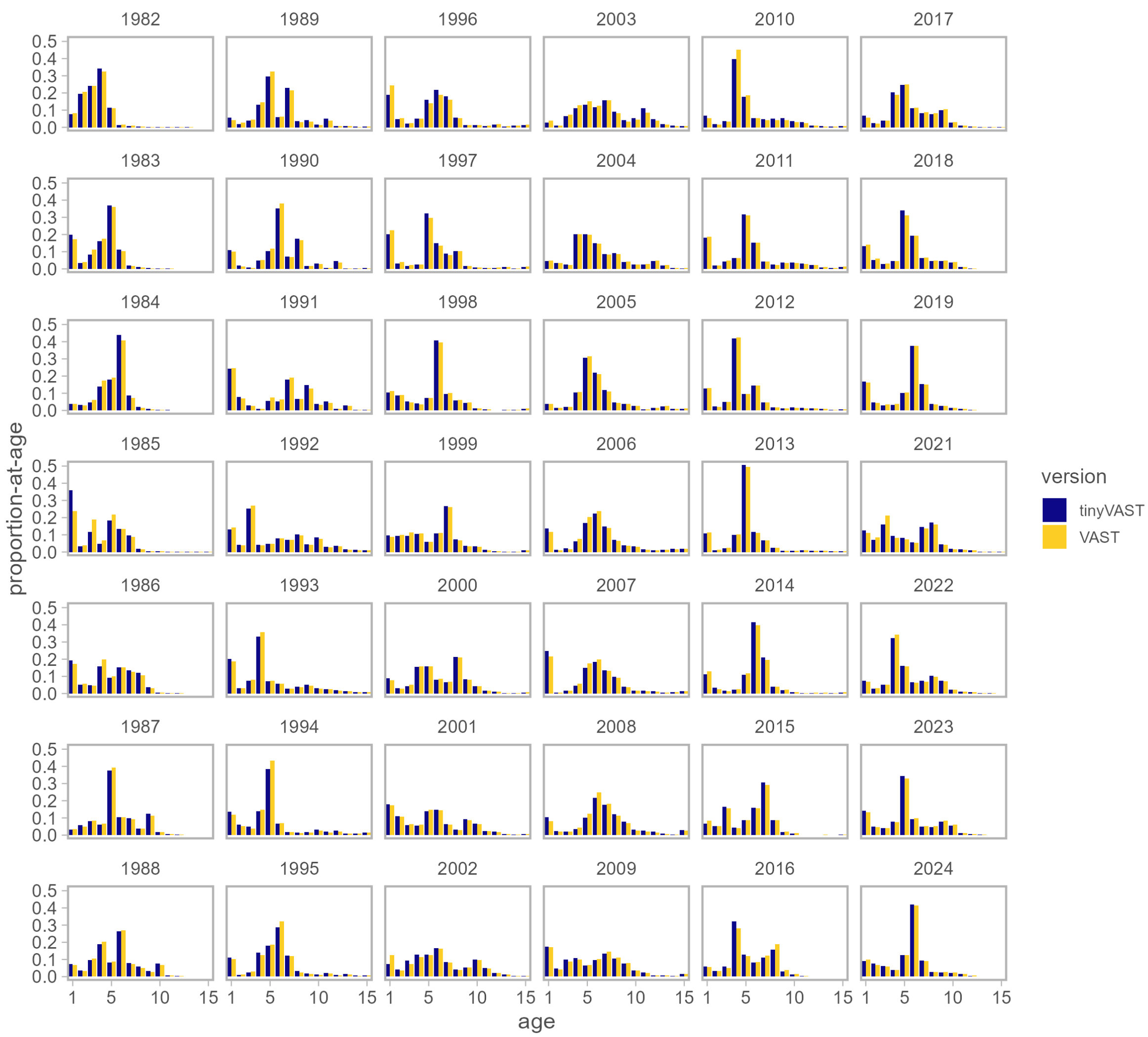

Figure 41: VAST Age Compare.

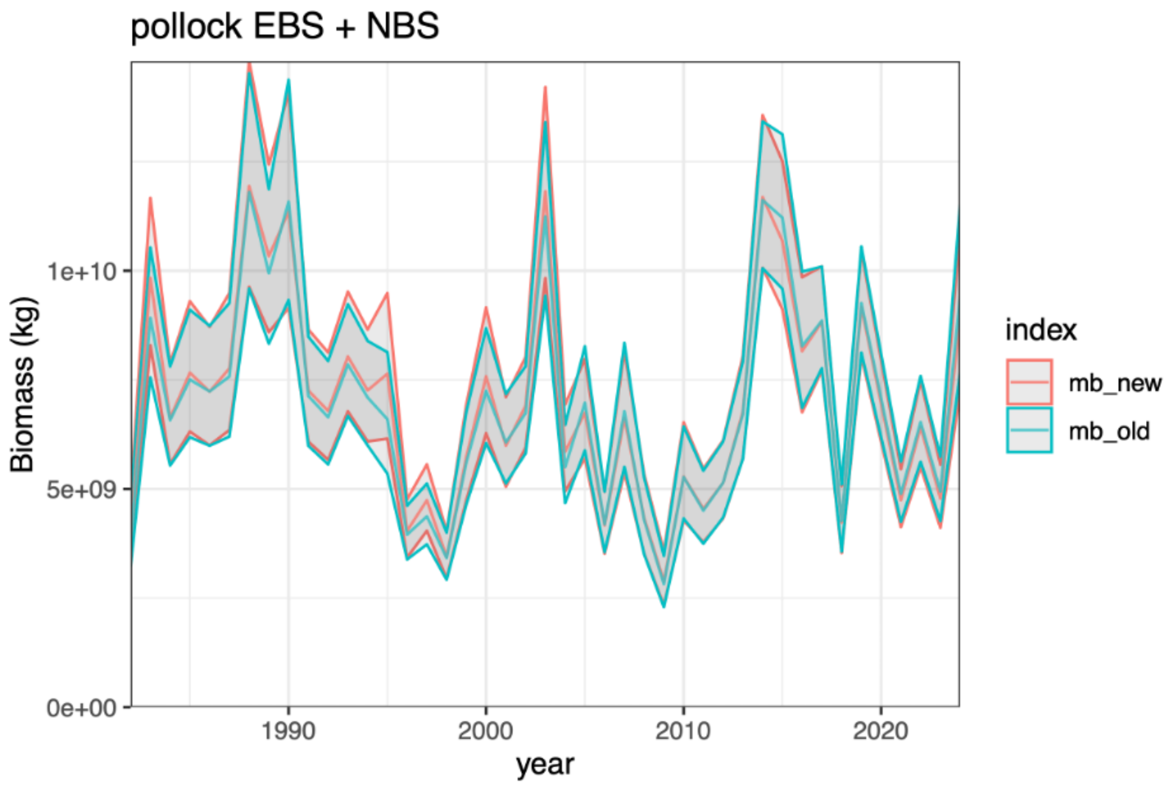

Figure 42: Vast Index Updated.